

Formula Sheet

O-Level (4024) & IGCSE (0580)

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Topic 1: Numbers

Natural Numbers(N):

Natural Numbers are counting numbers from 1,2,3,4,5,.....

$$N = \{1,2,3,4,5,.....\}$$

Whole Numbers (W):

Whole numbers are natural numbers including zero. They are 0,1,2,3,4,5,.....

$$W = \{0,1,2,3,4,5,.....\}$$

$$W = 0 + N$$

Prime number

Prime number is a positive natural **number** that has only two positive natural **number** divisors - one and itself. The opposite of **prime numbers** are composite **numbers**. A composite **number** is a positive natural **number** that has at least one positive divisor other than one or itself.

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97,
101, 103, 107, 109, 113, 127, 131, 137, 139, 149, 151, 157, 163, 167, 173, 179, 181, 191, 193, 197, 199, 211,
223, 227, 229, 233, 239, 241, 251, 257, 263, 269, 271, 277, 281, 283, 293, 307, 311, 313, 317, 331, 337,
347, 349, ...

Square Numbers

A square number, also called a perfect square, is a number of the form n^2 , where n is an integer. The square numbers for $n = 1, 2, 3, 4, 5, 6, 7 \dots$ are 0, 1, 4, 9, 16, 25, 36, 49, ...

Cube Numbers

A cube number is a number of the form n^3 , where n is an integer. The cube numbers for

$n = 1, 2, 3, 4, 5, 6, 7 \dots$ are 0, 1, 8, 27, 64, 125, 216, 343, ...

Positive Numbers:

Positive numbers are, 1,2 ,3 ,4 ,5,.....

Positive Numbers: $\{1, 2, 3, \dots\}$

Negative Numbers:

Negative numbers are,-3, -2, -1.

Negative integers: $\{\dots -3, -2, -1\}$

Integers (Z):

- Whole Numbers together with negative numbers.
- Integers are set containing the positive numbers, 1, 2, 3, 4,, and negative numbers,.....-3, -2, -1, together with zero.
- Zero is neither positive nor negative, but is both.
- In other words, Integers are defined as set of whole numbers and their opposites.
- $Z = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$

Decimals fractions have a FINITE NUMBER OF DIGITS after the **decimal** point. So, they are **terminating decimals**. However, there are situations when the digits after the **decimal** point DOES NOT END or they go on forever.

$3/5$ is terminating and 0.6 is a terminating decimal.

$17/8$ is terminating and 2.125 is a terminating decimal.

$1/4$ is terminating and 0.25 is a terminating decimal.

$27/24$ is terminating and 1.125 is a terminating decimal.

Non-Terminating Decimal Fractions

In some cases, a digit or a block of digits repeats itself in the **decimal** part. Such **decimals** are called **non-terminating** repeating **decimals** or pure recurring **decimals**. These **decimal** numbers are represented by putting a bar on the repeated part.

(a) $2.666\dots$ is a non-terminating repeating decimal and can be expressed as 2.6 .

(b) $0.141414\dots$ is a non-terminating repeating decimal and can be expressed as 0.14 .

Examples of mixed recurring decimals are:

(a) $3.1444\dots = 3.14$

(b) $8.12333\dots = 8.123$

(c) $7.3656565\dots = 7.365$

(c) $0.428571428571428571\dots$

Non-Recurring And Non-Terminating Decimal Fractions

decimal NEVER repeats and NEVER terminates, it cannot be represented as a ratio of two integers. So we call such number irrational (not a ratio). An example of this is the number

- An irrational number is a non-terminating and non-repeating decimal.
- All terminating and recurring decimals are RATIONAL NUMBERS.
- All non-terminating and non recurring decimals are IRRATIONAL NUMBERS.

$\sqrt{2} = 1.4142135623730950488016887242097\dots$

$\pi = 3.14159265358979\dots$

$0.123456789101112131415161718192021\dots$

Rational Numbers (Q):

- All numbers of the form a/b , where a and b are integers (but b cannot be zero)
- Rational numbers include fractions:
 - * Proper Fraction: Numbers smaller than 1 eg: $1/2$ or $3/4$
 - * Improper Fraction: Numbers greater than 1 eg: $5/2$
 - * Mixed Fraction: $2\frac{1}{2} = 5/2$
- Powers and square roots may be rational numbers if their standard form is a rational number.
- In rational numbers the denominator cannot be zero

Example:

2 can be expressed in the form of p/q as $2/1$

$-13/9 = -1.444.....$, $8^{-2} = 0.015625$

$(\sqrt{16})/3 = 4/3 = \pm 1.333...$ $\sqrt{4} = 2$, $1/2 = 0.5$ ----- Rational (terminates)

$2/3 = 0.6666666.....$ Rational (repeats) $5/11 = 0.454545.....$ Rational (repeats)

Irrational Numbers Q':

- Cannot be expressed as a ratio of integers.
- As decimals they never repeat or terminate.
- They go on for ever or infinity.
- Square root of all incomplete square or non-perfect square numbers

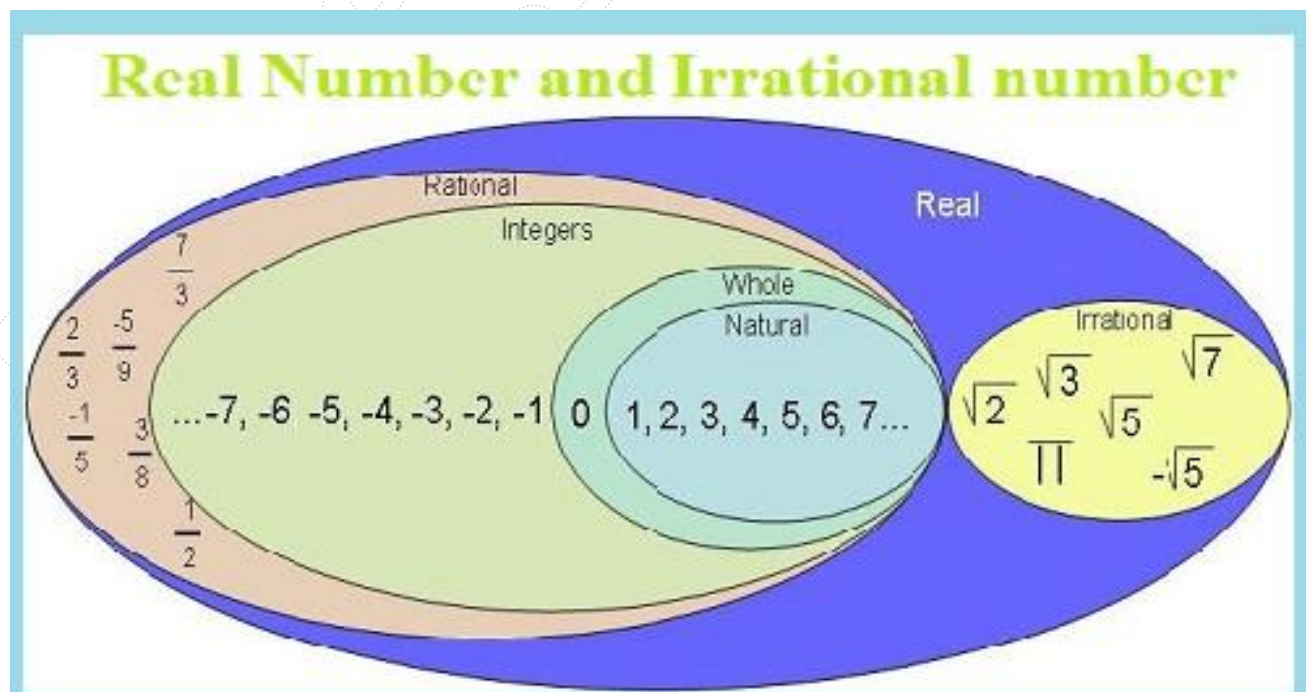
Example: $\sqrt{2}$, $\sqrt{3}$, $\sqrt{7}$, $\sqrt{8}$

square root of 2 = $\sqrt{2} = 1.41421356.....$ Irrational (never repeats or terminates)

$\pi = \pi = 22/7 = 3.14159265.....$ Irrational (never repeats or terminates)

Real Numbers R:

- Real Numbers are every number, irrational or rational.
- Any number that you can find on the number line.
- It is a number required to label any point on the number line; or it is a number that names the distance of any point from 0.
- **$R = Q + Q'$**
- Natural Numbers are Whole Numbers, which are Integers, which are Rational Numbers, which are Real Numbers.
- Irrational Numbers are Real Numbers, but not all Real Numbers are Irrational Numbers.



Topic 2. Significant figures & Decimal Places

Any number can be rounded off to a given number of significant figures (written s.f.) using the following rules.

- Count along to the number of significant figures required.
- Look at the next significant digit.
If it is smaller than 5, leave the 'significant' digits as they are.
If it is 5 or greater, add 1 to the last of the 'significant' digits.
- Restore the number to its correct size by filling with zeros if necessary.

e.g. Round 547.36 to 4, 3, 2, 1 significant figures.

$$547.36 = 547.4 \quad (4 \text{ s.f.})$$

$$547.36 = 547 \quad (3 \text{ s.f.})$$

$$547.36 = 550 \quad (2 \text{ s.f.})^*$$

$$547.36 = 500 \quad (1 \text{ s.f.})^*$$

NB You need to pad the numbers marked with an asterisk (*) with zeros in order to restore the numbers to their correct size.

Decimal places

Any number can be rounded to a given number of decimal places (written d.p.) using the following rules.

- Count along to the number of the decimal places required.
- Look at the digit in the next decimal place.
If it is smaller than 5, leave the preceding digits (the digits before it) as they are.
If it is 5 or greater, add 1 to the preceding digit.
- Restore the number by replacing any numbers to the left of the decimal point.

e.g. Round 19.3461 to 4, 3, 2, 1 decimal places.

$$19.3461 = 19.3461 \quad (4 \text{ d.p.})$$

$$19.3461 = 19.346 \quad (3 \text{ d.p.})$$

$$19.3461 = 19.35 \quad (2 \text{ d.p.})$$

$$19.3461 = 19.3 \quad (1 \text{ d.p.})$$

NB The numbers to the left of the decimal point are not affected by this rounding process as you are only concerned with decimal places.

Multiplying

e.g. Calculate 167×53

$$167 \times 53 = 167 \times (50 + 3) = 167 \times 50 + 167 \times 3 = 8350 + 501 = 8851$$

It is more usual to set this multiplication out like this.

$$\begin{array}{r} 167 \\ \times 53 \\ \hline 8350 \\ + 501 \\ \hline 8851 \end{array}$$

Multiplying by 50.

Multiplying by 3.

Adding.

Multiplying decimals

To multiply two decimals without using a calculator:

- ignore the decimal points and multiply the numbers
- add the number of digits after the decimal point in the numbers in the question
- position the decimal point so that the number of digits after the decimal point in the answer is the same as the total number of decimal places in the question.

e.g. Calculate 1.67×5.3

$$167 \times 53 = 8851 \text{ Ignoring the decimal points and multiplying the numbers.}$$

The number of digits after the decimal point in the numbers = $2 + 1 = 3$.

$1.67 \times 5.3 = 8.851$ Replacing the decimal point so that the number of digits after the decimal point in the answer is 3.

It is helpful to check that the answer is approximately correct i.e. 1.67×5.3 is approximately $2 \times 5 = 10$ so the answer of 8.851 looks correct.

Dividing

To divide by a two-digit number, proceed in exactly the same way as for any other division.

E.g. Calculate $513 \div 19$

$$\begin{array}{r} 27 \\ 19 \overline{) 513} \\ \underline{38} \downarrow \\ 133 \\ \underline{133} \\ 0 \end{array}$$

Topic 3(a): Rules of Divisibility

Remember these rules of divisibility.

A whole number is divisible by or can be divided exactly by:

Divisor	Condition	Example
2	If the Last digit is even (0, 2, 4, 6, 8)	28, 956, 8792, 75924
3	If the sum of its digits can be divided by 3	51, 123, 8742
4	if the last two digits can be divided by 4	$512 \rightarrow 12 \div 4 = 3 \rightarrow 512 \div 4 = 128$
5	If the last digit is 0 & 5	1450, 225,
6	if it is divisible by both 2 and 3	$216 \rightarrow 216 \div 2 = 108, 216 \rightarrow 216 \div 3 = 72 \rightarrow 216 \div 6 = 36$
7	If you double the last digit and subtract it from the rest of the number, and the answer is 0 or divisible by 7	343 $343 \rightarrow 343 - (2 \times 6)$ $\rightarrow 28$ As 28 is divisible by 7 so 343 is also divisible by 7.
8	If the last three digits are divisible by 8	
9	If the sum of the digits is a multiple of 9 (can be divided by 9)	
11	If the difference of the sum of alternative digits of a number is divisible by 11, then that number is divisible by 11 completely.	$\rightarrow 86416$, alternate digits (8, 4, 6) and the remaining alternate digits are (1 & 6). $\rightarrow 8 + 4 + 6 = 18$ and $1 + 6 = 7$. $18 - 7 = 11$, which is divisible by 11. Therefore 86416 is divisible by 11.
13	For any given number, to check if it is divisible by 13, we have to add four times of the last digit of the number to the remaining number and repeat the process until you get a two-digit number. Now check if that two-digit number is divisible by 13 or not. If it is divisible, then the given number is divisible by 13.	$2795 \rightarrow 279 + (5 \times 4)$ $\rightarrow 279 + (20)$ $\rightarrow 299$ $\rightarrow 29 + (9 \times 4)$ $\rightarrow 29 + 36$ $\rightarrow 65$ As 65 is divisible by 13, So 2795 is also divisible by 13

Topic 3(b): HCF & LCM

Highest Common Factor(HCF):

A multiple is obtained by multiplying a number by a natural number. The LCM of two or more numbers is the lowest multiple found in all the sets of multiples.

HCF = Common factors with least Power

$$168 = 2^3 \times 3 \times 7$$

$$180 = 2^2 \times 3^2 \times 5$$

$$\text{HCF} = 2^2 \times 3 = 12$$

Least Common Multiple (LCM)

A factor of a number divides into it exactly. The HCF of two or more numbers is the highest factor found in all the sets of factors

LCM = (Common factors with highest Power) × (Non- Common factors)

$$168 = 2^3 \times 3 \times 7$$

$$180 = 2^2 \times 3^2 \times 5$$

$$\text{LCM} = 2^3 \times 3^2 \times 5 \times 7 = 2520$$

Topic 4: BODMAS

Many people use the letters **BODMAS** to remember the order of operations. The letters stand for:

B Brackets

Of (Sometimes, 'I' for 'indices' is used instead of 'O' for 'of')

Divide **M**ultiply

Add **S**ub

Topic 5: Fraction

To **multiply** two fractions together, multiply the numerators together and multiply the denominators together.

$$2\frac{1}{3} \times 1\frac{1}{2}$$

$$= \frac{7}{3} \times \frac{3}{2}$$

$$= \frac{21}{6} = \frac{7}{2} = 3\frac{1}{2}$$

To **divide** two fractions, multiply the first fraction by the reciprocal of the second fraction. This means simply that the divide sign is swapped for a multiply sign, and the second fraction is flipped upside down.

$$\frac{3}{5} \times \frac{3}{2} = \frac{3 \times 3}{5 \times 2} = \frac{9}{10}$$

For example, it is not possible to add $\frac{3}{5}$ and $\frac{1}{4}$ without changing each fraction so that they have the same bottom number.

We can use *equivalent fractions* to rewrite each fraction with 20 as the denominator:

$$\frac{3}{5} = \frac{12}{20} \text{ and } \frac{1}{4} = \frac{5}{20}$$

Now we can see that $\frac{3}{5} + \frac{1}{4} = \frac{12}{20} + \frac{5}{20} = \frac{17}{20}$

Two fractions can only be added or subtracted if they have the same denominator

$$\frac{3}{5} + \frac{1}{5} = \frac{4}{5}$$

If fractions do not have the same denominator, you need to find *equivalent* fractions which do

$$\frac{3}{5} + \frac{1}{4} = \frac{12}{20} + \frac{5}{20} = \frac{17}{20}$$

These fractions do not have the same denominator, so the first step is to change them so that they do. In this case, we can use 20 as the common denominator.

$$\frac{1}{4} + \frac{2}{5} = \frac{5}{20} + \frac{8}{20} = \frac{5+8}{20} = \frac{13}{20}$$

In this case we can use a common denominator of 12.

$$\frac{2}{3} - \frac{1}{4} = \frac{8}{12} - \frac{3}{12} = \frac{8-3}{12} = \frac{5}{12}$$

Topic 6. Limit & Accuracy (Upper & Lower Bounds)

The value 9.5 cm is called the **lower bound** as it is the lowest value which would be rounded to 10cm while 10.5 cm is called the **upper bound**.

A rectangle measures 10 cm by 6 cm where each measurement is given to the nearest cm. Write down an interval approximation for the area of the rectangle.

Lower bound (minimum area) = $9.5 \times 5.5 = 52.25\text{cm}^2$

Upper bound (maximum area) = $10.5 \times 6.5 = 68.25\text{cm}^2$

Upper Bound = Maximum Value

Lower Bound = Minimum Value

	Lower bound	Upper bound
8000 Km given to the correct nearest 1000 Km	7500 Km	8500 Km
250Kg given to the correct nearest 10 Kg	245 Kg	255 Kg
4.7g given to the correct nearest tenth of a gram	4.65 g	4.75 g
27.54 sec given to the correct nearest hundredth a sec	27.535 sec	27.545 sec

$$\text{Max } (p + q) = p_{\text{max}} + q_{\text{max}}$$

$$\text{Min } (p + q) = p_{\text{min}} + q_{\text{min}}$$

$$\text{Max } (p \times q) = p_{\text{max}} \times q_{\text{max}}$$

$$\text{Min } (p \times q) = p_{\text{min}} \times q_{\text{min}}$$

$$\text{Max } (p - q) = p_{\text{max}} - q_{\text{min}}$$

$$\text{Min } (p - q) = p_{\text{min}} - q_{\text{max}}$$

$$\text{Max } (p \div q) = p_{\text{max}} \div q_{\text{min}}$$

$$\text{Min } (p \div q) = p_{\text{min}} \div q_{\text{max}}$$

$$\text{Upper Bound } (p + q) = p_{\text{upper}} + q_{\text{upper}}$$

$$\text{Lower Bound } (p + q) = p_{\text{lower}} + q_{\text{lower}}$$

$$\text{Upper Bound } (p \times q) = p_{\text{upper}} \times q_{\text{upper}}$$

$$\text{Lower Bound } (p \times q) = p_{\text{lower}} \times q_{\text{lower}}$$

$$\text{Upper Bound } (p - q) = p_{\text{upper}} - q_{\text{lower}}$$

$$\text{Lower Bound } (p - q) = p_{\text{lower}} - q_{\text{upper}}$$

$$\text{Upper Bound } (p \div q) = p_{\text{upper}} \div q_{\text{lower}}$$

$$\text{Lower Bound } (p \div q) = p_{\text{lower}} \div q_{\text{upper}}$$

Topic 7. Laws of Indices

(i) $a^m \times a^n = a^{m+n}$

(ii) $a^m \div a^n = a^{m-n}$

(iii) $(a^m)^n = a^{mn}$

(iv) $a^m \times b^m = (a \times b)^m$

(v) $a^m \div b^m = (a \div b)^m$

Where 'm' and 'n' are positive integers, $m > n$, $a \neq 0$, $b \neq 0$

(vi) $a^0 = 1$ Where 'a' is any real number and $a \neq 0$

(viii) $\frac{1}{a^n} = a^{-n}$, $a \neq 0$ and 'n' is positive integer.

(viii) $a^{\frac{1}{n}} = \sqrt[n]{a}$ $a > 0$ and 'n' is positive integer.

(ix) $a^{\frac{m}{n}} = \sqrt[n]{a^m}$ 'm' and 'n' is positive integers and $a > 0$.

(x) $\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n$ Where 'a' and 'b' are any real numbers, $a \neq 0$ & $b \neq 0$, and 'n' is positive integer.

Topic 8. Algebra

Mathematics Important Formulae

$(a + b)^2 = a^2 + 2ab + b^2$

$(a - b)^2 = a^2 - 2ab + b^2$

$(a + b)(a - b) = a^2 - b^2$

Quadratic Formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Standard form of Quadratic $ax^2 + bx + c = 0$

Topic 9: Patterns & Sequences

Linear sequences

the n th term = first term + $(n - 1) \times 1$ st difference

Quadratic sequences

n th term = first term + $(n - 1) \times 1$ st difference + $\frac{1}{2} (n - 1)(n - 2) \times 2$ nd difference

Special sequences

The following are special sequences of numbers that you should be able to recognise.

	Name of Sequence	n th term
1, 4, 9, 16, 25, ...	square numbers	n^2
1, 8, 27, 64, 125, ...	cube numbers	n^3
1, 3, 6, 10, 15, ...	triangle numbers	$(n + 1)(n + 2) \div 2$
2, 3, 5, 7, 11, 13, 17, ...	prime numbers	
1, 1, 2, 3, 5, 8, 13, 21, ...	Fibonacci sequence	

Constant Difference Sequences (Linear Sequence)

This is when the difference between terms is always the same.

e.g. 1, 4, 7, 10, ... This has a difference which is always 3.

How do you find the formula for the ' n th' term?

Well, the three times table has the formula ' $3n$ ' and the terms in this sequence are two less than the terms in the three times table so the formula is ' $3n - 2$ '.

You can always find the ' n th term' by using this formula:

$$n\text{th term} = a + (n - 1)d$$

Where **d** is the difference between the terms, **a** is the first term and **n** is the term number.

Obviously these are more difficult but once again we can use a formula!

$$n\text{th term} = a + (n - 1)d + \frac{1}{2}(n - 1)(n - 2)c$$

This time there is a letter **c** which stands for the **second difference** (or the difference between the differences and **d** is just the difference between the **first two** numbers).

Topic 10: Matrices

Types of Matrices:

Row Matrix:

$$\begin{bmatrix} 1 & 3 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 2 & -3 & 6 \end{bmatrix}$$

Coloumn Matrix:

$$\begin{bmatrix} 5 \\ 7 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 10 \\ 8 \\ -6 \end{bmatrix}$$

Special Matrix:

$$\begin{bmatrix} 2 & 5 \\ -3 & 9 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 1 & 0 & 1 \\ 3 & -4 & 2 \\ 2 & -1 & 5 \end{bmatrix}$$

Null Matrix (0):

Null Matrix is that matrix, that only contains number 0 in it.

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \quad \text{Or} \quad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Identity (or unit) Matrix (I):

The elements in the diagonal are one's only.

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Writing the order of Matrices:

Order = Number of Rows * Number of Columns

Example:

$$\begin{bmatrix} 1 & 0 \\ 3 & -4 \\ 2 & -1 \end{bmatrix} \quad \text{Order} = 3 \times 2$$

$$\begin{bmatrix} 2 & 5 & 2 \\ -3 & 9 & 7 \end{bmatrix} \quad \text{Order} = 2 \times 3$$

Addition and Subtraction of Matrices:

Matrices of the same order are added (or subtracted) by adding (or subtracting) the corresponding elements in each matrix.

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$$

$$\text{Adding } A + B = \begin{bmatrix} a+e & b+f \\ c+g & d+h \end{bmatrix} ; \quad \text{Subtracting } A - B = \begin{bmatrix} a-e & b-f \\ c-g & d-h \end{bmatrix}$$

Rules:

$A+B = B+A$: A and B can change position when adding.

$(A+B) + C = A + (B+C)$: order of operation bracket first.

$A - B \neq B - A$: A and B should not change positions when subtracting.

Example:

$$A = \begin{bmatrix} 1 & 3 \\ -2 & 8 \end{bmatrix} \quad \text{and } B = \begin{bmatrix} -4 & 6 \\ 0 & 7 \end{bmatrix}$$

$$A + B = \begin{bmatrix} 1 & 3 \\ -2 & 8 \end{bmatrix} + \begin{bmatrix} -4 & 6 \\ 0 & 7 \end{bmatrix}$$

$$= \begin{bmatrix} -3 & 9 \\ -2 & 15 \end{bmatrix}$$

$$= (B + A)$$

Scalar Multiplication of a Matrix by a real number:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$kA = k \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$= \begin{bmatrix} ka & kb \\ kc & kd \end{bmatrix}$$

$$= k(A + B) = KA + KB \text{ (Ans)}$$

$$A = \begin{bmatrix} 5 & 15 \\ -10 & 40 \end{bmatrix}$$

$$\frac{1}{5} \begin{bmatrix} 5 & 15 \\ -10 & 40 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 \\ -2 & 8 \end{bmatrix} \text{ (Ans)}$$

Example:

Equal Matrix:

If two matrices A and B are of the same order and their corresponding elements are equal, then $A = B$.

Example:

$$3 \begin{bmatrix} 2x \\ y \end{bmatrix} + 3 \begin{bmatrix} x \\ 3y \end{bmatrix} = \begin{bmatrix} 18 \\ 36 \end{bmatrix}$$

$$\begin{bmatrix} 6x \\ 3y \end{bmatrix} + \begin{bmatrix} 3x \\ 9y \end{bmatrix} = \begin{bmatrix} 18 \\ 36 \end{bmatrix}$$

$$\begin{bmatrix} 9x \\ 12y \end{bmatrix} = \begin{bmatrix} 18 \\ 36 \end{bmatrix}$$

$$9x = 18$$

$$12y = 36$$

$$x = 2 \text{ (Ans)}$$

$$y = 3 \text{ (Ans)}$$

Multiplication of Two or More Matrices:

Matrices can only be multiplied only if they are compatible. They are compatible when the number of rows of the second matrix is the same as the number of columns of the first matrix.

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$$

$$AB = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} e & f \\ g & h \end{bmatrix}$$

$$BA = \begin{bmatrix} e & f \\ g & h \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\begin{bmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{bmatrix}$$

$$\begin{bmatrix} ea + fc & eb + fd \\ ga + hc & gb + hd \end{bmatrix}$$

Rules:

In general $AB \neq BA$: A and B should not change positions

(AB)C = A(BC): If 3 or more matrices you can choose whichever 2 to multiply first.

Example:

Determinant Inverse Of Matrix:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Determinant A = Det A = |A| = ad - bc

(If Det A = 0 then A is singular and (If Det A = 0 Det A = 0 then A is singular

$$A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Remember:

- If Determinant = 0 then the matrix has no inverse.
 - Multiplying by the inverse of the inverse of a matrix gives the same result as dividing by the matrix.
- E.g.

$$\text{If } AB = C$$

$$A^{-1}AB = A^{-1}C$$

$$B = A^{-1}C$$

Example of Inverse of Matrix:

- If Determinant = 0 then matrix has no inverse.
- Multiplying by the inverse of the inverse of a matrix gives the same result as dividing by the matrix.E.g.

$$\text{Find } A^{-1} \text{ of } A = \begin{bmatrix} -6 & 7 \\ 4 & 3 \end{bmatrix}$$

$$A^{-1} = \frac{1}{(-6 \times 3) - (+4 \times 7)} \begin{bmatrix} 3 & -7 \\ 4 & -6 \end{bmatrix}$$

$$= \frac{1}{-10} \begin{bmatrix} 3 & -7 \\ 4 & -6 \end{bmatrix}$$

$$= \begin{bmatrix} 3/10 & -7/10 \\ 4/10 & -6/10 \end{bmatrix}$$

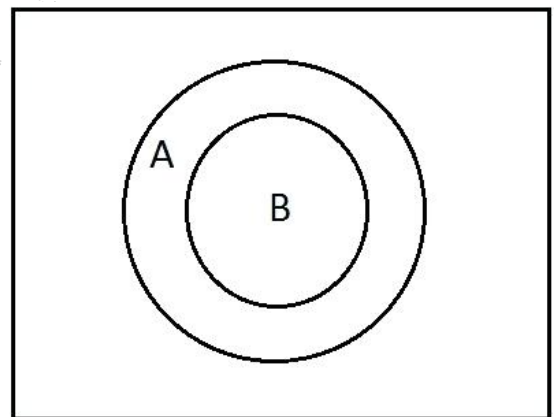
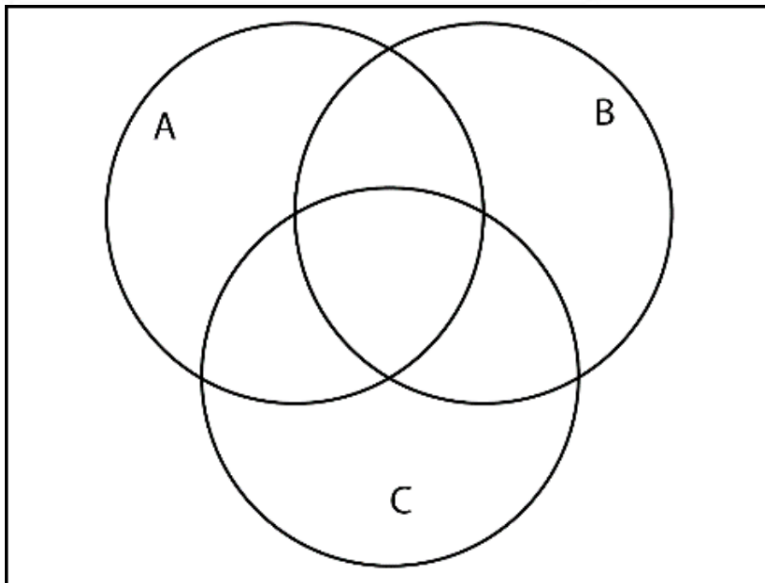
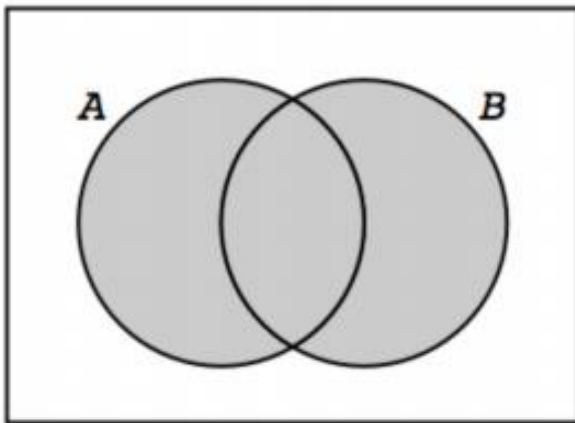
$$= \begin{bmatrix} 3/10 & -7/10 \\ 2/5 & -3/5 \end{bmatrix} \text{ (Ans)}$$

Topic 11: Set Notation

SETS:

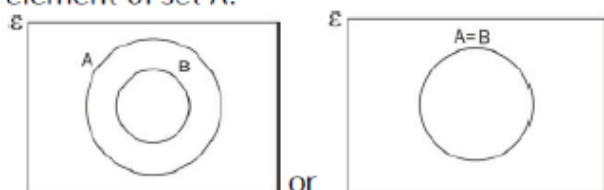
Notations

- ξ = universal set
- $\cup$ (union) = all the elements
- $\cap$ (intersection) = common elements
- $\emptyset$ or $\{ \}$ = empty set
- $\in$ = belongs to
- $\notin$ = does not belongs to
- $\subseteq$ = Subset
- A' = compliment of A (i.e. the elements of ξ - the elements of A)
- $n(A)$ = the number of elements in A.
- De Morgan's Laws: $(A \cup B)' = (A' \cap B')$
 $(A \cap B)' = (A' \cup B')$



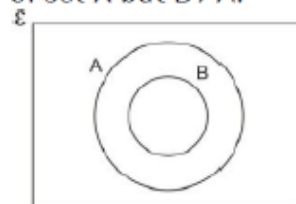
Subset $\subseteq$

$B \subseteq A$ means every elements of set B is also an element of set A.



Proper subset $\subset$

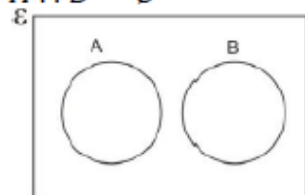
$B \subset A$ means every element of B is an element of set A but $B \neq A$.



Disjoint sets

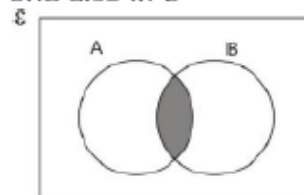
Disjoint set do not have any element in common. If A and B are disjoint sets, then

$$A \cap B = \emptyset$$



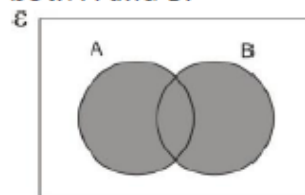
Intersection $\cap$

$A \cap B$ is the set of elements which are in A and also in B



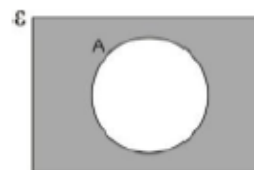
Union $\cup$

$A \cup B$ is the set of elements in either A, B or both A and B.



Complement

The complement of A, written as A' refers to the elements in $\mathcal{E}$ but not in A.



Topic 12: (Co-ordinate Geometry)

Distance Formula

Two points $A(x_1, y_1)$ and $B(x_2, y_2)$

$$|AB| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} \quad \text{or} \quad |AB| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

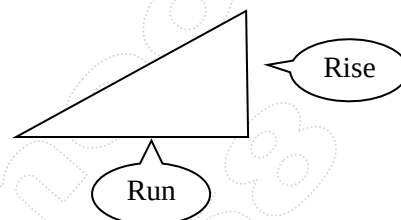
Mid Point

Two points $A(x_1, y_1)$ and $B(x_2, y_2)$

$$\text{Mid point of AB} = C \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Gradient of Line

$$\text{Gradient} = \frac{\text{Rise}}{\text{Run}}$$



Two points $A(x_1, y_1)$ and $B(x_2, y_2)$

$$\text{Gradient of Line AB} = m = \frac{y_2 - y_1}{x_2 - x_1} \quad \text{or} \quad m = \frac{y_1 - y_2}{x_1 - x_2}$$

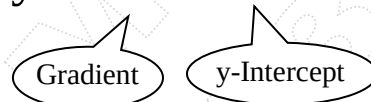
Gradient of Parallel Lines are same.

Product of Gradient of Perpendicular Lines = -1

Or Gradient of Perpendicular line to AB = -1 / Gradient of AB

Equation of Line

$$y = mx + c$$



Alternative way: When one point $P(x_1, y_1)$ and gradient m are given. $(y - y_1) = m(x - x_1)$

x-intercept

For x- intercept put $y = 0$ in the given equation.

x- intercept (....., 0)

y-intercept

For y- intercept put $x = 0$ in the given equation.

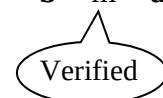
y- intercept (0,)

Any point on given line

If any point $A(a, b)$ on the given line $y = mx + c$ then it should satisfied the given equation.

$$b = m \times a + c$$

Converse is also valid



Area of triangle

An triangle ABC have vertices $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$

$$\text{Area of } \Delta ABC = \frac{1}{2} \{ (x_1 \times y_2 - x_2 \times y_1) + (x_2 \times y_3 - x_3 \times y_2) + (x_3 \times y_1 - x_1 \times y_3) \}$$

OR

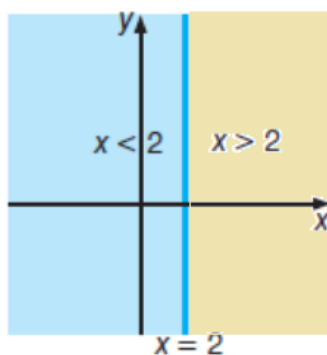
$$\text{Area of } \Delta ABC = \frac{1}{2} \{ (x_1 \times y_2 + x_2 \times y_3 + x_3 \times y_1) - (x_2 \times y_1 + x_3 \times y_2 + x_1 \times y_3) \}$$

For Finding Co-ordinate 4th Vertex

In Parallelogram, Rectangle, Rhombus and square if there co-ordinates of their vertices $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ are given then we can find co-ordinate of fourth vertex $D(a, b)$.

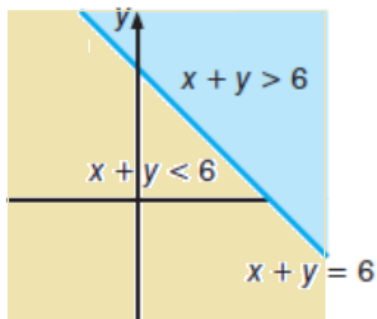
$$x_1 + x_3 = x_2 + a \quad \& \quad y_1 + y_3 = y_2 + b$$

Topic 13: Graphing inequalities

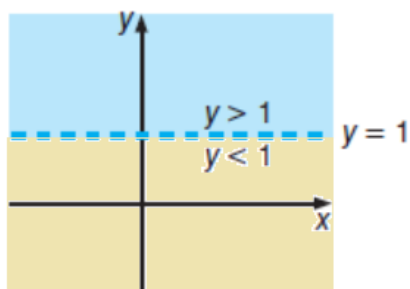


Vertical Line
Required region $x \geq 2$ includes the line $x = 2$.

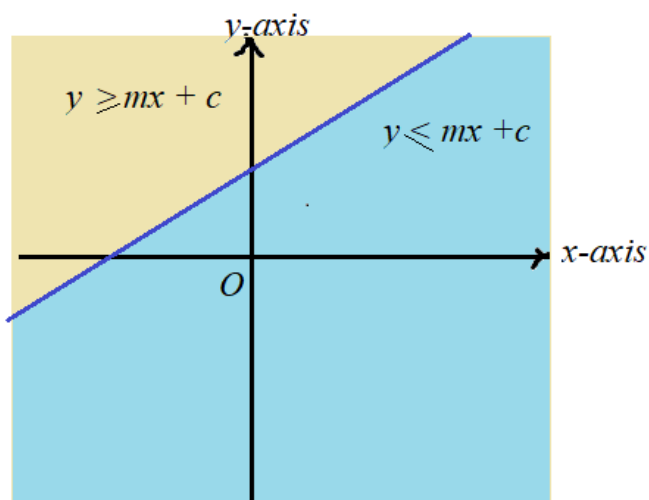
Special Case
Required region $x + y \leq 6$ includes the line $x + y = 6$.



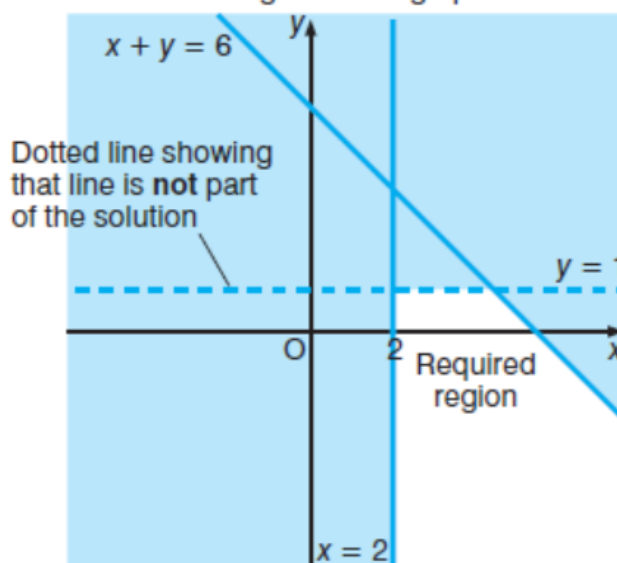
$ax + by < c$ or $ax + by \leq c$



Horizontal Line
Required region $y < 1$ does not include the line $y = 1$



Now combining the three graphs.



- Always keep co-efficient of x as Positive.
- If face of inequality is open towards x then region is right to the line
- Otherwise left side of the line (If face of inequality is open opposite to x)

Topic 14: Percentage

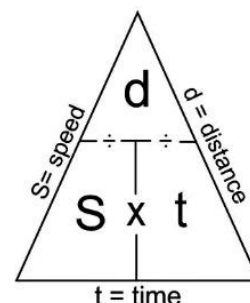
Profit:	= Selling Price – Cost Price
Profit %:	= Profit / Cost Price x 100
Loss:	= Cost Price – Selling Price
Loss %:	= Loss / Cost Price x 100
Discount:	= Marked Price – Sale Price
Discount %:	= Discount / Marked Price x 100
Percentage Increase:	= Increase / Original Cost x 100%
Simple Interest	I = PRT/100
Compound Interest	CI = P (1 + $\frac{r}{100}$)ⁿ – P

Total Amount (Principle + Compound Interest) = $P (1 + \frac{r}{100})^n$

Amount: = **Principle + Interest**

Topic 15: Time & Speed

Departure Time:	= Arrival Time – Journey Time
Arrival Time:	= Departure Time + Journey Time
Journey Time:	= Arrival Time – Departure Time
Speed:	= Distance / Time Taken
Time Taken:	= Distance / Speed
Distance:	= Speed x Time Taken



Topic 16: Measure

Unit Conversion

Length:

$$\begin{aligned} 1 \text{ cm} &= 10 \text{ mm} \\ 1 \text{ m} &= 100 \text{ cm} \\ 1 \text{ Km} &= 1000 \text{ m} = 100,000 \text{ cm} \end{aligned}$$

Volume:

$$\begin{aligned} 1 \text{ m}^3 &= 1000 \text{ Litres} \\ 1 \text{ litre} &= 1000 \text{ ml} = 1000 \text{ cm}^3 \end{aligned}$$

Time:

$$\begin{aligned} 1 \text{ minute} &= 60 \text{ sec} \\ 1 \text{ hour} &= 60 \text{ minutes} = 3600 \text{ second} \end{aligned}$$

Speed

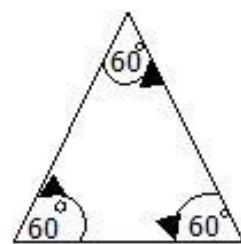
$$1 \text{ metre/second (m/sec)} = 3.6 \text{ Kilometre / hour (Km/h)}$$

$$1 \text{ Kilometre / hour (Km/h)} = (1/3.6) \text{ metre/second (m/sec)}$$

Topic 17: Geometrical Terms

Equilateral Triangles

Triangles with all three sides equal in length and all three angles equal in magnitude, are called equilateral triangles. Since the angles in a triangle sum to 180° and the size of each angle is the same in an equilateral triangle, the angles are all 60° .



Equal sides
Equal angles

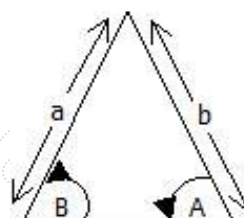
Equilateral Triangle

Isosceles Triangles

Isosceles triangles are triangles with two sides equal in length and angles equal in magnitude.

$$\text{Base Angle} = (180^\circ - \text{Vertex Angle})/2$$

$$\text{Vertex Angle} = 180^\circ - 2 \times \text{Base Angle}$$



$$\angle A = \angle B$$

$$a = b \quad \text{two}$$

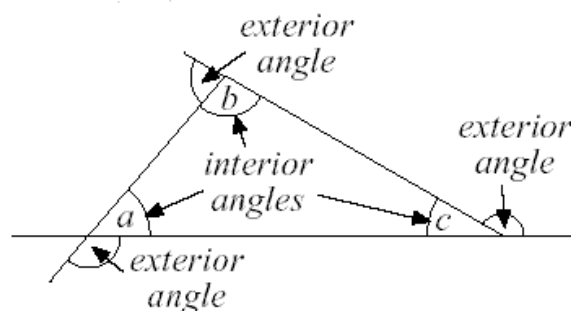
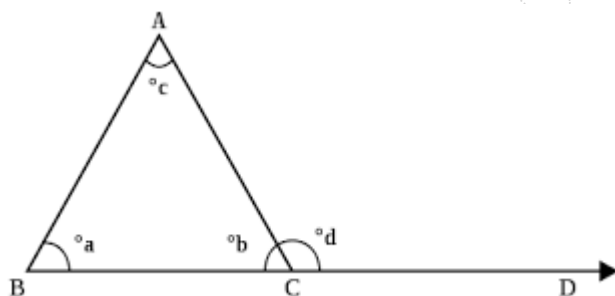
Isosceles Triangle

External Angle

External Angle = Sum of opposite internal angles

$$a + c = d$$

$$A + B = \text{ACD}$$



Types And Properties Of Quadrilaterals

Quadrilaterals are any four-sided figure, joined by four straight lines, that is, Quadrilaterals are four-sided polygons.

Properties of Quadrilaterals:

- Four sides.
- Four vertices (corners).
- Interior angles sum to 360° .
- Exterior angles sum to 360° .

Things to Know:

- Diagonals are line segments that join two opposite vertices (corners).

Therefore:

M and O – are opposite vertices (Corners)

M and N – are called adjacent vertices since they are on the same sides of figure.

- Two sides are adjacent, if they share a common vertex.

RS are adjacent sides, since they share the common vertex, R.

Note, two sides are opposite sides if they are directly across from each other.

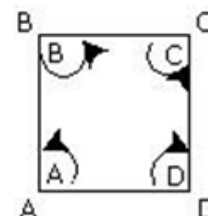
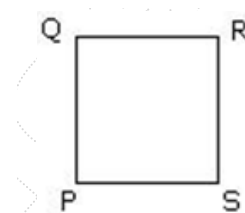
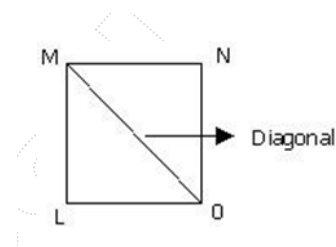
There: QR and PS are opposite sides.

The sides QR and

- Two angles are adjacent, if they share a common side.

Angles B and C Are adjacent.

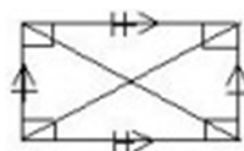
Note, two angles A and C are opposite if they are directly across from each other. That is, angle A and C are opposite.



Rectangle

Note:

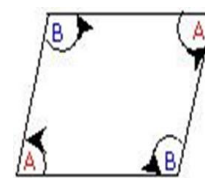
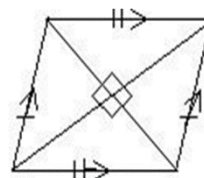
- All four angles are right angles.
- Opposite sides are equal and parallel
- Diagonals are equal in length and bisect each other.
- Diagonals form two pairs of congruent triangles (revise congruent triangles)



□ means "right angle"
 $\overleftrightarrow{H}$ & $\overleftrightarrow{+}$ show equals sides

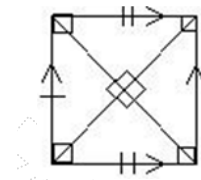
Rhombus

- All four sides are equal.
- Opposite sides are parallel.
- Opposite angles are equal.
- Four congruent triangles formed by diagonals.



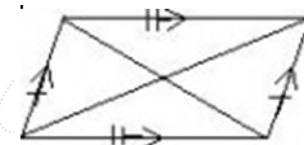
Square

- All four sides are equal.
- All four angles are right angles.
- Opposite sides are parallel.
- Diagonals bisect each other at right angles.
- Diagonals are of equal length.
- Four congruent triangles formed by diagonals.

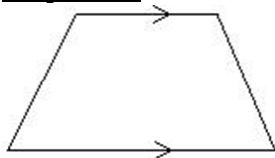


Parallelogram

- Opposite sides are equal.
- Opposite sides are parallel.
- Opposite angles are equal.
- Diagonals bisect each other to form two pairs of congruent triangles.



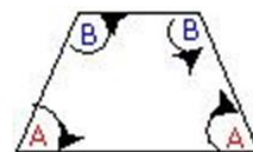
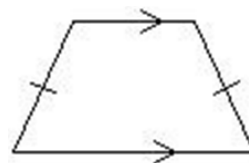
Trapezium



- One pair of opposite sides parallel.

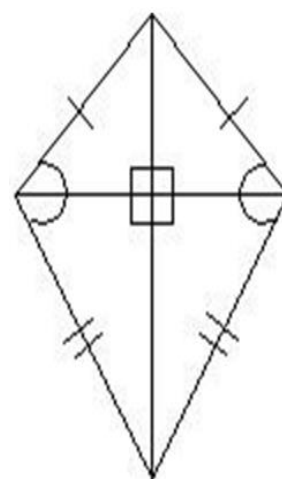
Isosceles Trapezium

- One pair of opposite sides parallel.
- The pair of non-parallel sides are equal in length.
- Adjacent angles on the same parallel sides are equal.



Kite

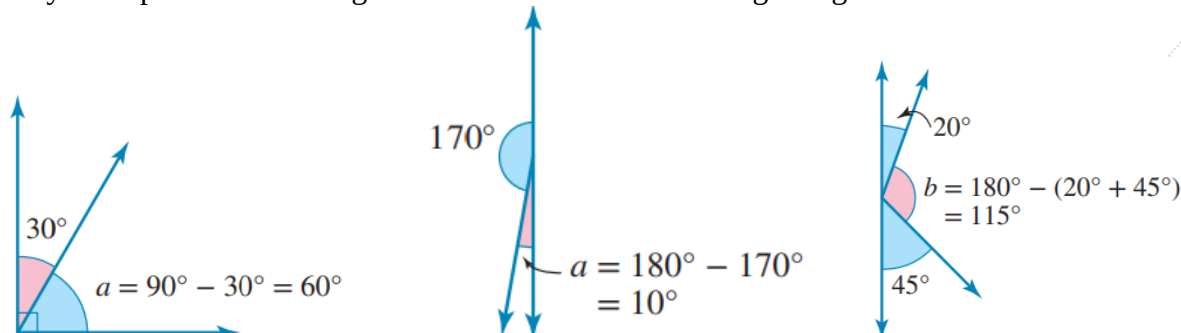
- Two pairs of equal in length adjacent sides.
- Diagonals intersect at right angles.
- One pair of equal opposite angles.
- One diagonals is bisected by the other.
- Two pairs of congruent triangles formed.



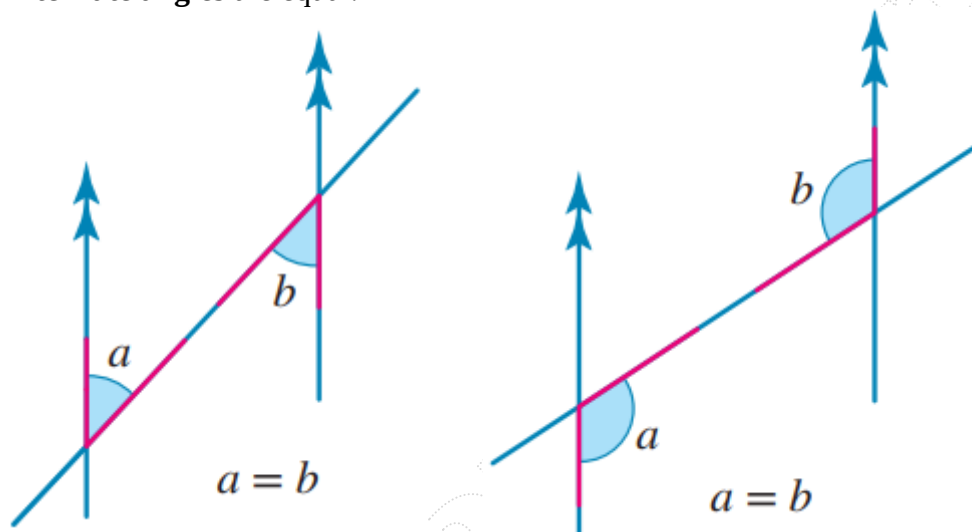
Topic 18: Angles, Angle properties of polygons

Angle

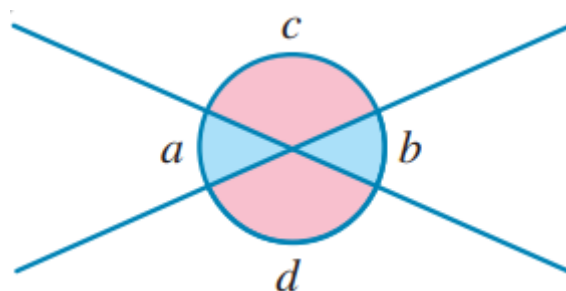
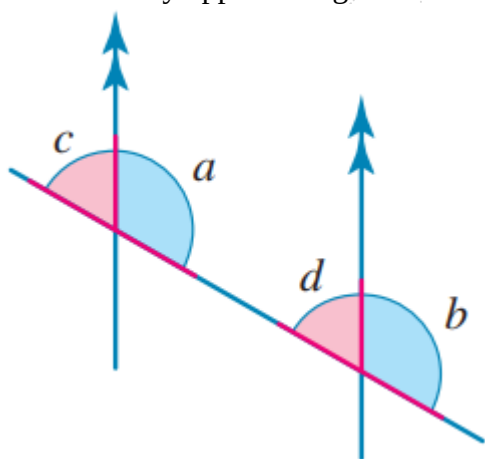
Two or more angles are **complementary** if they add up to 90° . Two or more angles are **supplementary** if they add up to 180° . An angle of 180° is also called a straight angle.



For **alternate angles** to exist we need a minimum of one pair of parallel lines and one transverse line. **Alternate angles** are equal.



Other types of angles to be considered are corresponding angles, co-interior angles, triangles in a semicircle and vertically opposite angles.



Corresponding angles are equal::

$$a = b$$

$$c = d$$

Co-interior angles are supplementary:

$$a + d = 180^\circ$$

Vertically opposite angles are equal:

$$a = b$$

$$c = d$$

Acute angle

less than 90°



Right angle

$= 90^\circ$



Obtuse angle

between 90° and 180°



Straight line

$= 180^\circ$



Reflex angle

greater than 180°



Complete turn

$= 360^\circ$



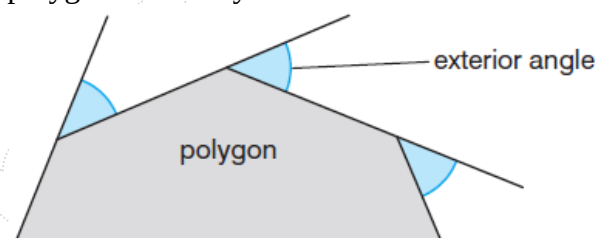
Exterior angles

The exterior angle of a polygon is found by continuing the side of the polygon externally.

A **regular** polygon has all sides equal and all angles equal.

A **convex** polygon has no interior angle greater than 180° .

A **concave** (or **re-entrant**) polygon has at least one interior angle greater than 180° .



Angle properties of Polygon

Regular Polygon:

All sides and angles are equal.

Exterior Angles + Interior Angle = 180°

Sum of Interior Angles of any Polygon = $(n - 2) \times 180^\circ$

Each Interior Angle of regular Polygon = $\frac{(n-2)}{n} \times 180^\circ$

Sum of Exterior Angles of any Polygon = 360°

Each exterior Angle of regular Polygon = $\frac{360^\circ}{n}$

Size of Exterior Angles

Interior and Exterior angles are measured on the same line, that is, they add up to 180° .

Therefore, the size of an exterior angle = $180^\circ - \text{Interior angle}$.

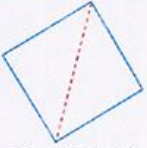
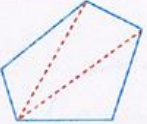
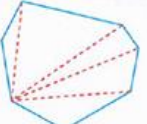
For example, the size of the external angle of the pentagon above is:

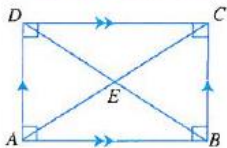
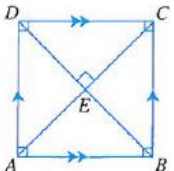
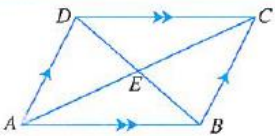
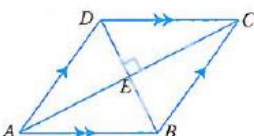
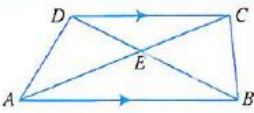
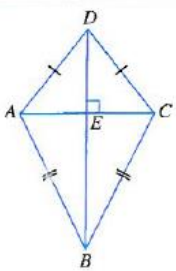
Since, interior angle = 108°

Then, exterior angle = $180^\circ - \text{Interior angle}$

Below is a list of the names and the number of sides, of some of the most popular polygons.

Name of Polygon	Number of Sides
Equilateral Triangle	3
Quadrilateral	4
Pentagon	5
Hexagon	6
Heptagon	7
Octagon	8
Nonagon	9
Decagon	10

Polygon	Number of sides	Number of Triangle(s) formed	Sum of Interior Angles	Size of Interior Angles Regular Polygon	Sum of Exterior Angles	Size of Exterior Angles Regular Polygon
 Triangle	3	1	$1 \times 180^\circ = (3 - 2) \times 180^\circ$	$1 \times 180^\circ = (3 - 2) \times 180^\circ$	360°	$\frac{360^\circ}{3} = 60^\circ$
 Quadrilateral	4	2	$2 \times 180^\circ = (4 - 2) \times 180^\circ$	$2 \times 180^\circ = (4 - 2) \times 180^\circ$	360°	$\frac{360^\circ}{4} = 90^\circ$
 Pentagon						
 Hexagon						
 Heptagon						
 Octagon						
 Nonagon			360°			
 Decagon						
n -gon (Polygon)	n	$(n - 2)$	$(n - 2) 180^\circ$	$\frac{(n - 2) 180^\circ}{n}$	360°	$\frac{360^\circ}{n}$

Quadrilateral	Parallel sides	Equal sides	Interior angles	Diagonals
 Rectangle	There are two pairs of parallel sides.	Opposite sides are equal in length.	All four angles are right angles.	- The two diagonals are equal in length. - Diagonals bisect each other, i.e. $AE = EC$ and $BE = ED$.
 Square	There are two pairs of parallel sides.	All four sides are of the same length.	All four angles are right angles.	- The two diagonals are equal in length. - Diagonals bisect each other at right angles, i.e. $\hat{AEB} = \hat{BEC} = \hat{AED} = \hat{CED} = 90^\circ$. - Diagonals bisect the interior angles, e.g. $\hat{BAC} = \hat{CAD} = 45^\circ$ and $\hat{ABD} = \hat{CBD} = 45^\circ$.
 Parallelogram	There are two pairs of parallel sides.	Opposite sides are equal in length.	Opposite angles are equal, i.e. $\hat{ADC} = \hat{ABC}$ and $\hat{BAD} = \hat{BCD}$.	Diagonals bisect each other, i.e. $AE = EC$ and $BE = ED$.
 Rhombus	There are two pairs of parallel sides.	All four sides are of the same length.	Opposite angles are equal, i.e. $\hat{ADC} = \hat{ABC}$ and $\hat{BAD} = \hat{BCD}$.	- Diagonals bisect each other at right angles, i.e. $\hat{AEB} = \hat{BEC} = \hat{AED} = \hat{CED} = 90^\circ$. - Diagonals bisect the interior angles, e.g. $\hat{BAC} = \hat{CAD}$ and $\hat{ABD} = \hat{CBD}$.
 Trapezium	There is at least one pair of parallel sides.			
 Kite	The opposite sides may be parallel.	There are two pairs of equal adjacent sides.		- Diagonals cut each other at right angles, i.e. $\hat{AEB} = \hat{BEC} = \hat{AED} = \hat{CED} = 90^\circ$. - One diagonal bisects the interior angles, i.e. $\hat{ADB} = \hat{CDB}$ and $\hat{ABD} = \hat{CBD}$.

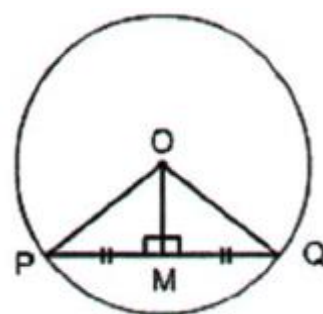
Topic 19: Symmetrical and Angle properties of Circles

Property 1 (Perpendicular Bisecting Chord)

A straight line drawn from the centre of a circle to bisect a chord, which is not a diameter, is perpendicular to the chord. Conversely, the perpendicular to a chord, drawn from the centre of a circle, bisects the chord.

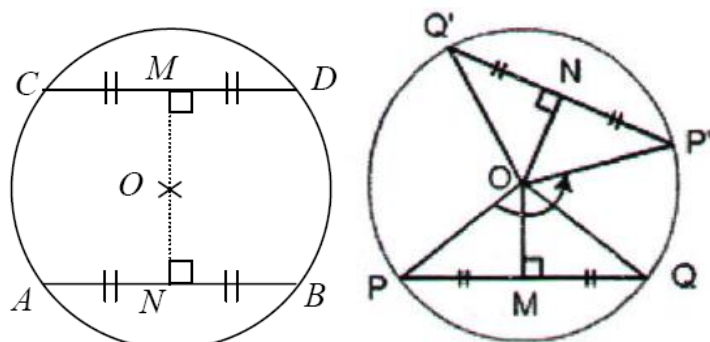
($\perp$ from centre bisects chord)

$$PQ \perp OM$$



Property 2 (Equal Chords)

In equal circles or in the same circle, equal chords are **equidistant** from the centres or centre. Conversely, chords which are equidistant from the centre are equal.



$$PQ = P'Q' \text{ and } OM = ON$$

Angles in the same segment

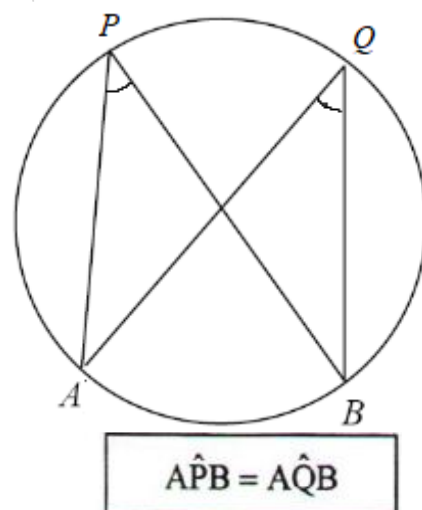
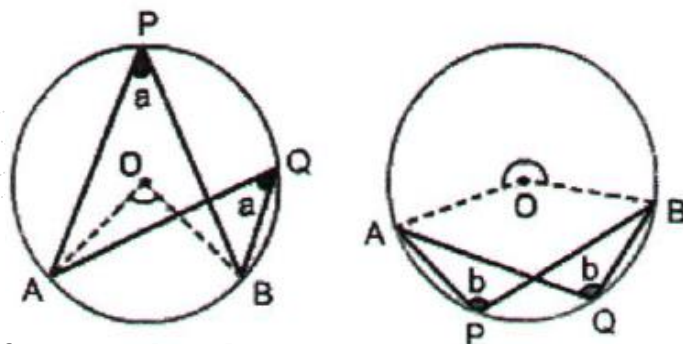
Angle APB and Angle AQB are said to be angles subtended by the same arc AB or by the chord AB.

Property 3 (Angles in Same Segment)

-Angles in the same segment or two equal segments of a circle are equal.

($\angle$ s in the same segment)

$$\text{Angle XAY} = \text{Angle XBY}$$

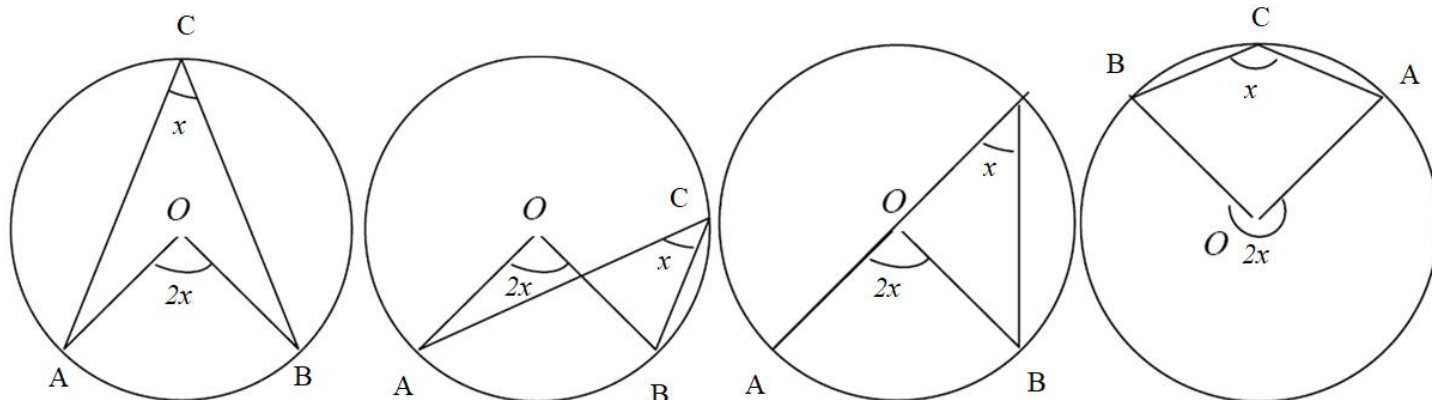


$$\hat{A}PB = \hat{A}QB$$

Property 4 (Angle at Circumference and Angle at Centre)

The angle subtended by an arc at the centre of a circle is twice that subtended at the circumference of the circle.

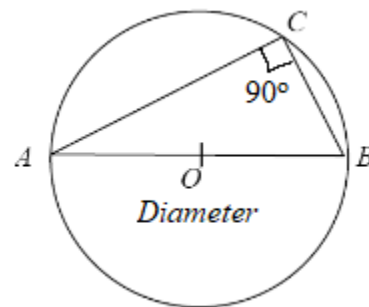
($\angle$ s at centre = $2\angle$ at $\odot^{ce}$)



Property 5: (Angle in Semicircle)

An angle subtended by the diameter of a circle at the circumference of the circle is a **right angle**.

Angle $ACB = 90^\circ$ (rt. $\angle$ in a semicircle)



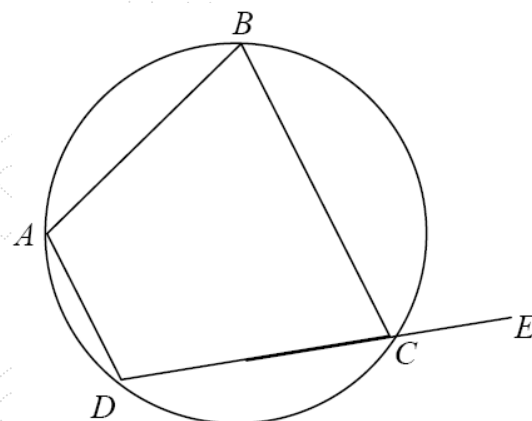
Cyclic Quadrilaterals

A quadrilateral with its four vertices lying on the circumference of a circle is called a cyclic quadrilateral

Property 6:

Angles in opposite segment are supplementary (The sum of the angles in opposite segment add up to 180°)

Abbreviated to ($\angle$ s in opp. Segment are supp.)



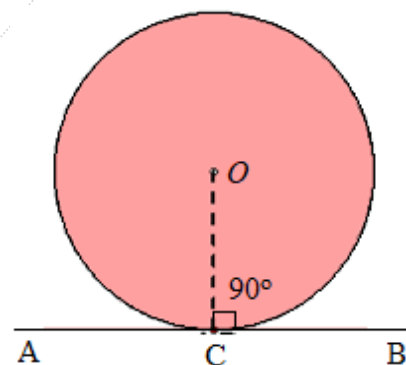
Property 7:

Tangent is perpendicular (Tangent $\perp$ Radius)

A straight line and a circle have only one point of contact, then that line is called tangent.

$OC \perp AB$

$\angle OCA = \angle OCB$



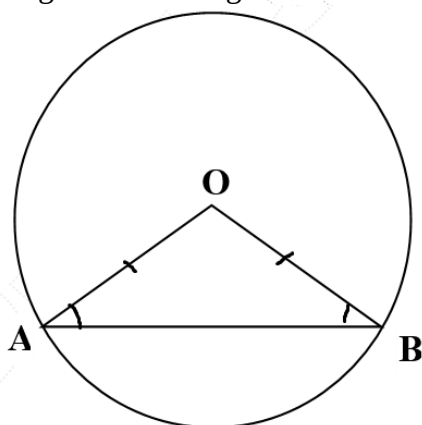
Property 8:

If two sides of a triangle are radius of the circle then that triangle is an Isosceles triangle.

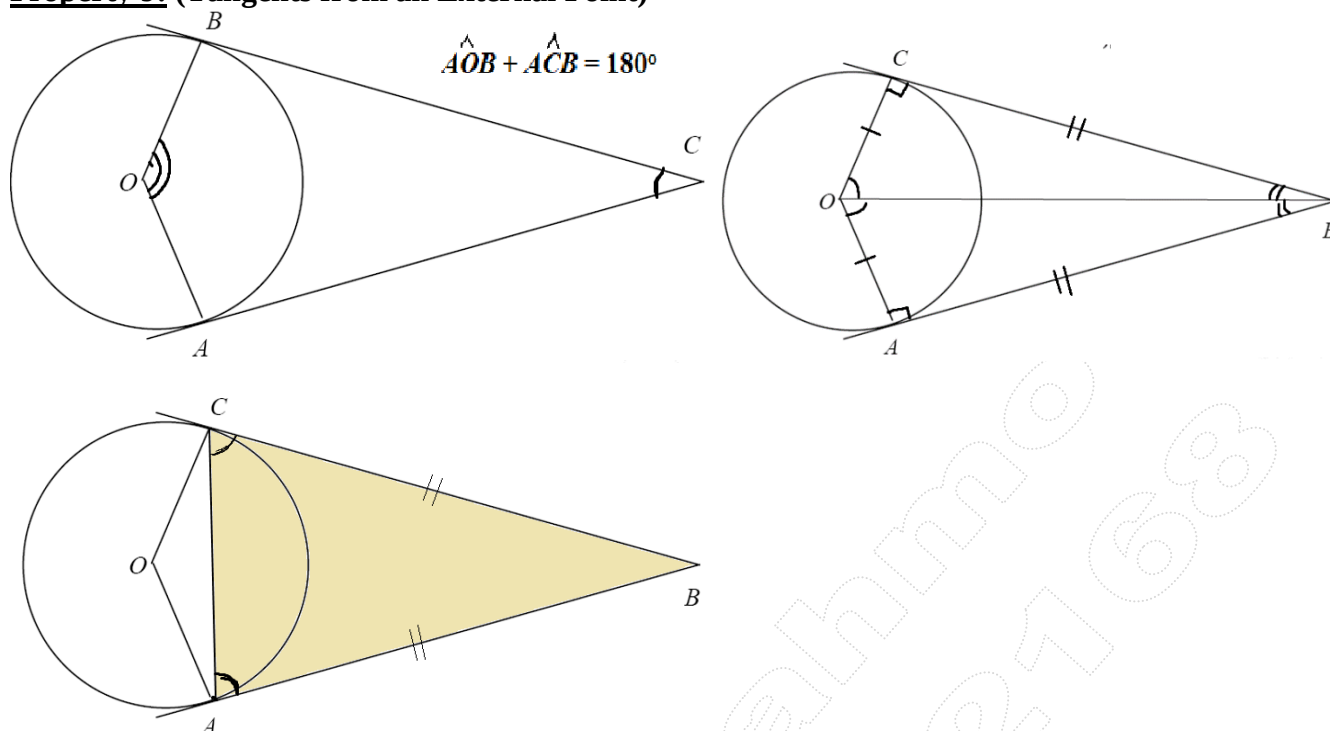
In triangle $\triangle OAB$,

$OA = OB$,

Angle $OAB = \text{Angle } OBA$



Property 9: (Tangents from an External Point)

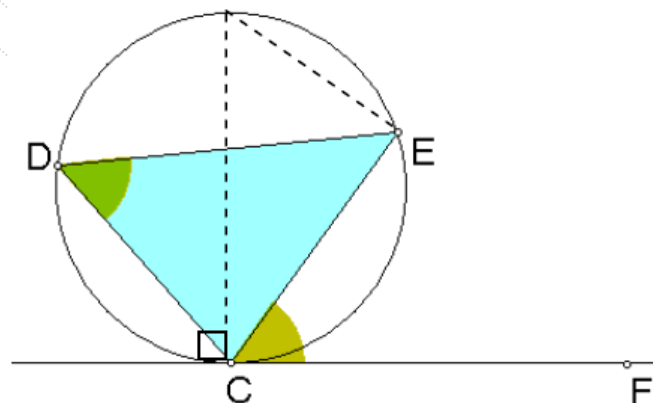


- (i) Tangents drawn to a circle from an external point are equal;
- (ii) The tangents subtend equal angles at the centre;
- (iii) The line joining the external point to the centre of the circle bisects the angle between the tangents.

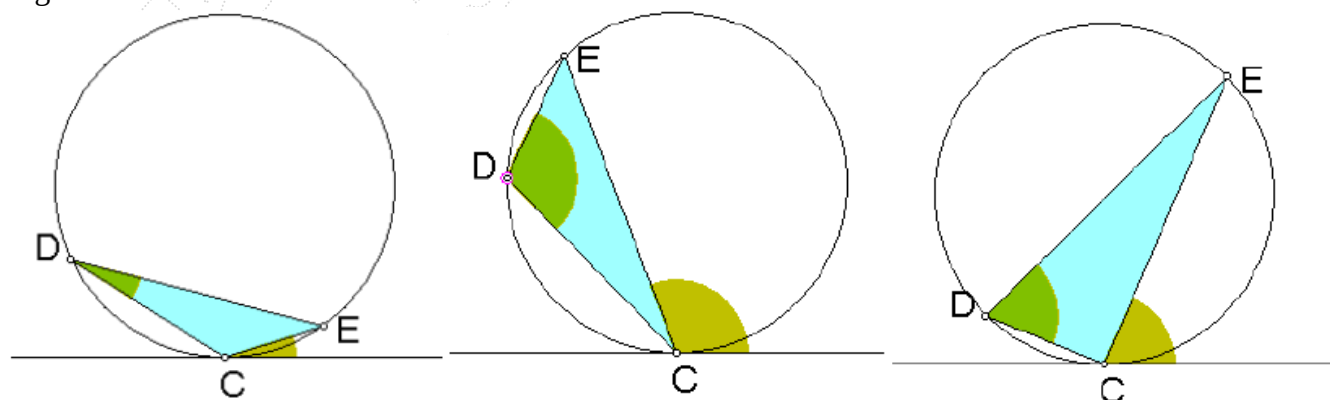
Property 10: (Alternate Segment Theorem)

Angle between chord and tangent line is same as angle subtended by the same chord in circle

“The angle between the tangent and chord at the point of contact is equal to the angle in the **alternate segment**”.



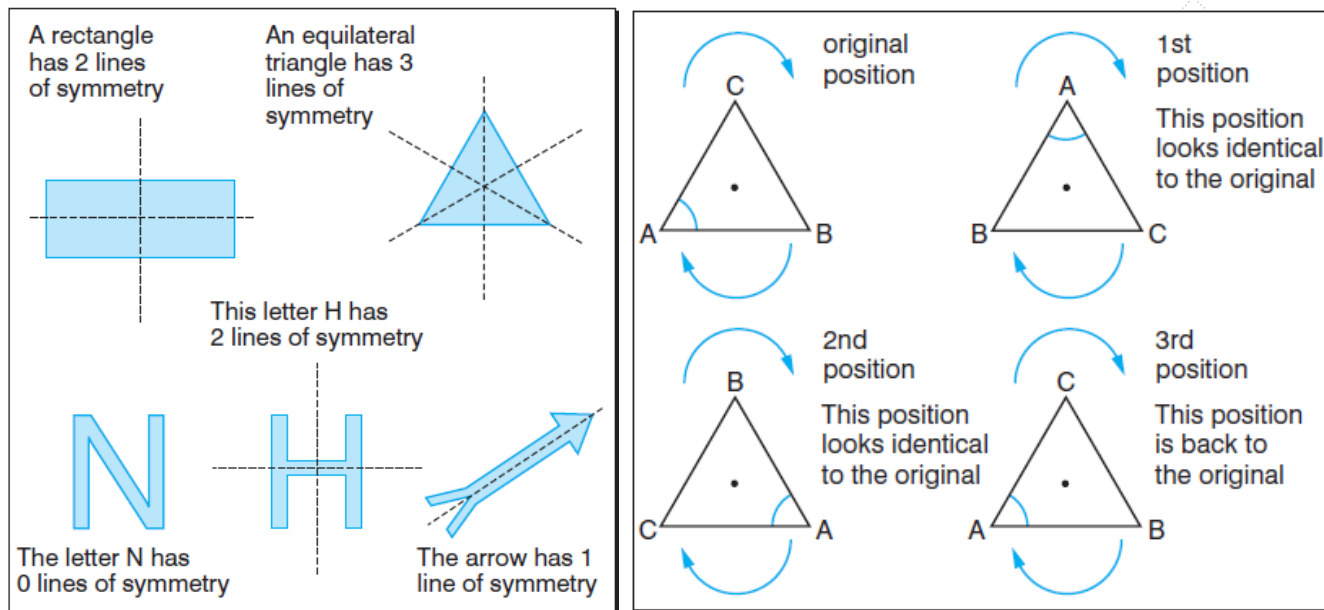
Here are some other possible situations of the alternate segment theorem.



Topic 20: Symmetry

Line symmetry

When a shape can be folded so that one half fits exactly over the other half, the shape is symmetrical and the fold line is called a **line of symmetry**.



Rotational symmetry

When a shape can be rotated about its centre to fit exactly over its original position, the shape has **rotational symmetry**.

The number of **different** positions gives the **order** of rotational symmetry. An equilateral triangle has rotational symmetry of order 3.

Topic 21: Trigonometry

Pythagoras Theorem:

Sine θ :

Cosine θ :

Tangent θ :

$0^\circ < \theta < 90^\circ$

$$(\text{Hypotenuse})^2 = (\text{Base})^2 + (\text{Perpendicular})^2$$

Perpendicular / Hypotenuse

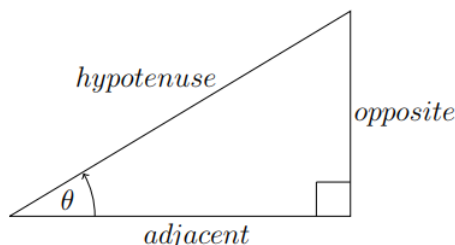
Base / Hypotenuse

Perpendicular / Base

SOH $\sin \theta = \frac{\text{opp}}{\text{hyp}}$

CAH $\cos \theta = \frac{\text{adj}}{\text{hyp}}$

TOA $\tan \theta = \frac{\text{opp}}{\text{adj}}$



Law of Sines

Solving Oblique Triangles using sine: AAS, ASA, SSA, SSS, SAS

Law of Cosines

Cosine: SAS, SSS

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} \quad \text{or} \quad \frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

Standard Form

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$b^2 = a^2 + c^2 - 2ac \cos B$$

$$c^2 = b^2 + a^2 - 2ab \cos C$$

Alternative Form

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

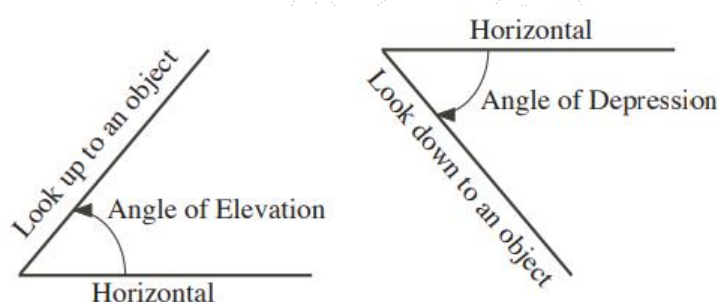
$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Finding the Area of non-90° Triangles (Oblique Δ)

$$\text{area} = \frac{1}{2}bc \sin A = \frac{1}{2}ab \sin C = \frac{1}{2}ac \sin B$$

Angle of Elevation & Depression

The angle of elevation is the angle above the horizontal that an observer must up look to see an object that is higher than the observer. The angle of depression is the angle below the horizontal that an observer must look down to see an object that is lower than the observer.



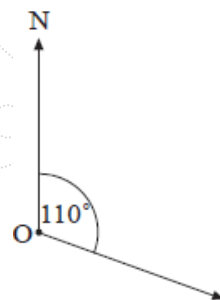
Bearings

A bearing is an angle, measured clockwise from the north direction. Below, the bearing of B from A is 025 degrees (note 3 figures are always given). The bearing of A from B is 205 degrees.

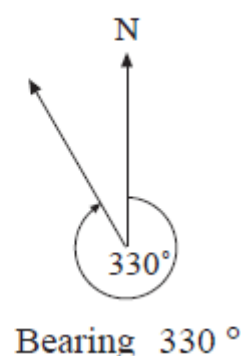
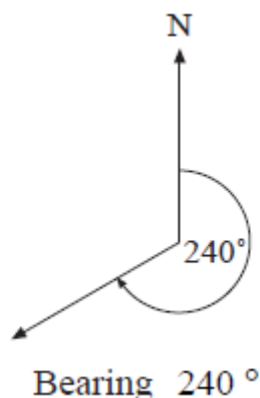
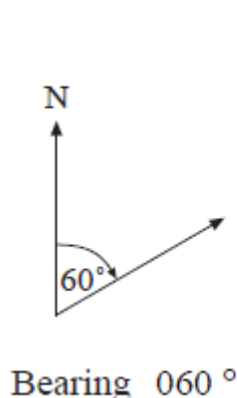
Bearing

Bearings are a measure of direction, with north taken as a reference. If you are traveling north, your bearing is 000°.

If you walk from O in the direction shown in the diagram, you are walking on a bearing of 110°.



Bearings are always measured *clockwise from north*, and are given as three figures, for example:



Back or Reciprocal Bearing

A ship sails from A to B on a bearing of 060°. On what bearing must it sail if it is to return from B to A?

The diagram shows the journey from A to B.

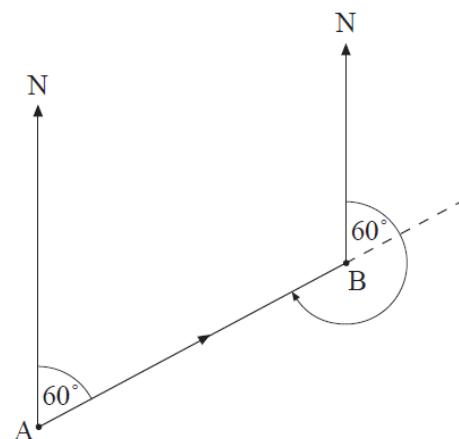
Extending the line of the journey allows an angle of 60° to be marked at B.

The diagram shows the journey from A to B.

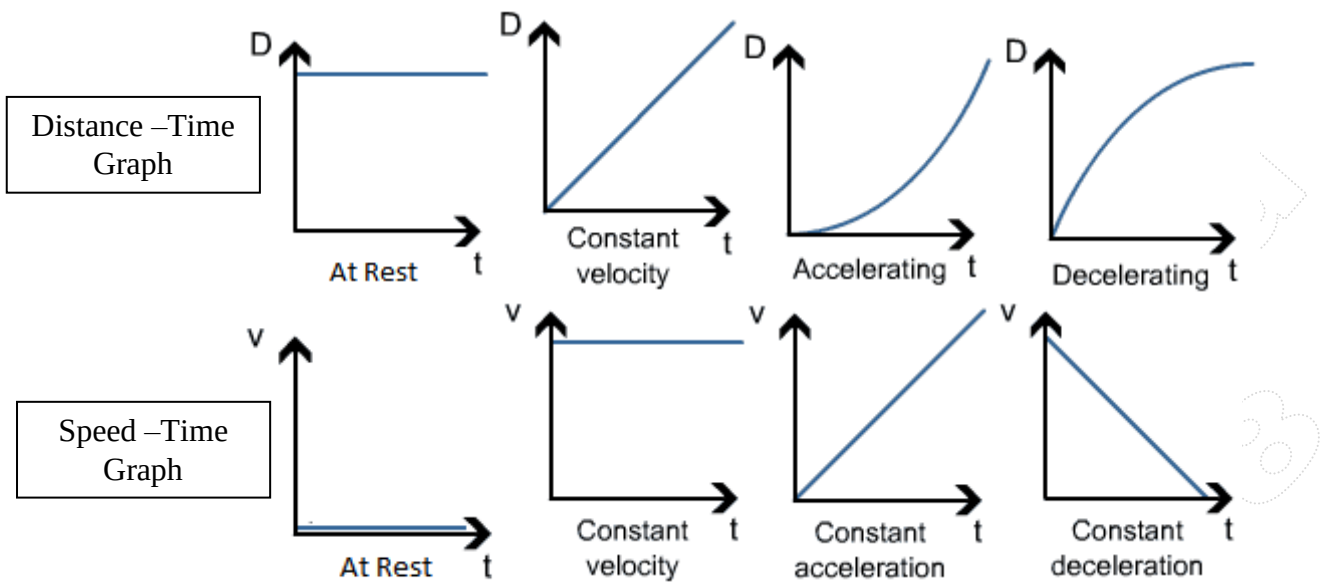
Extending the line of the journey allows angle of 60° to be marked at B.

$$\begin{aligned} \text{Bearing of A from B} &= 060^\circ + 180^\circ \\ &= 240^\circ \end{aligned}$$

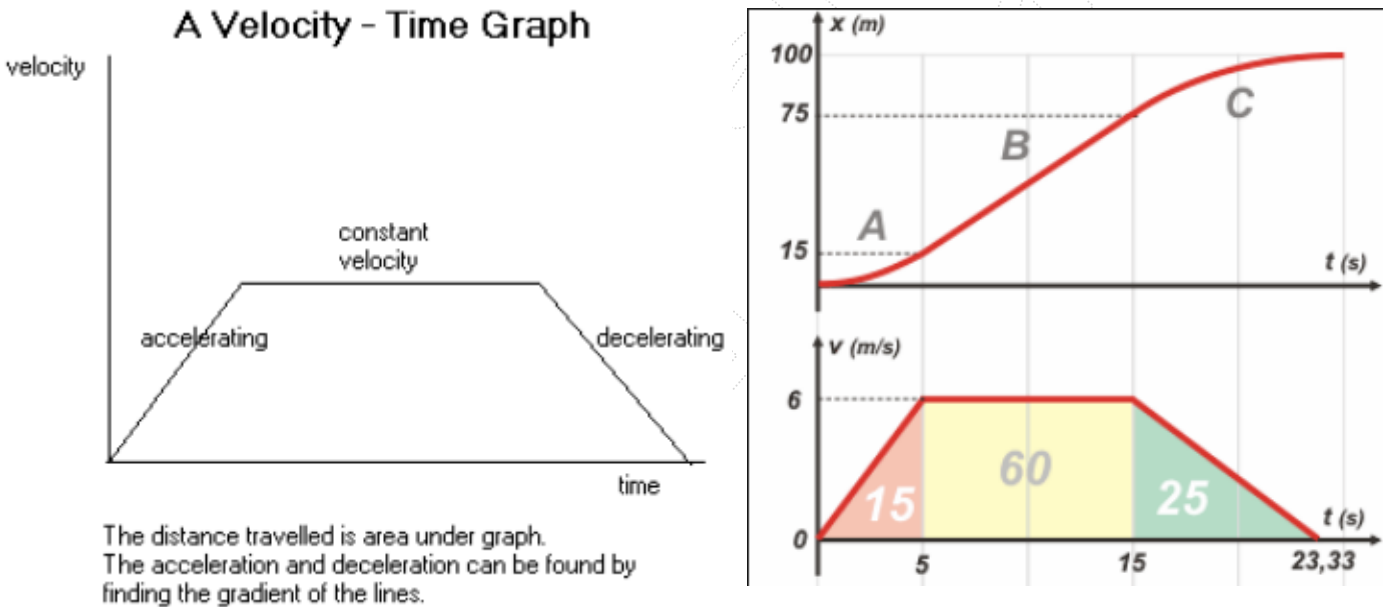
and this is called a *back bearing* or a *reciprocal bearing*.



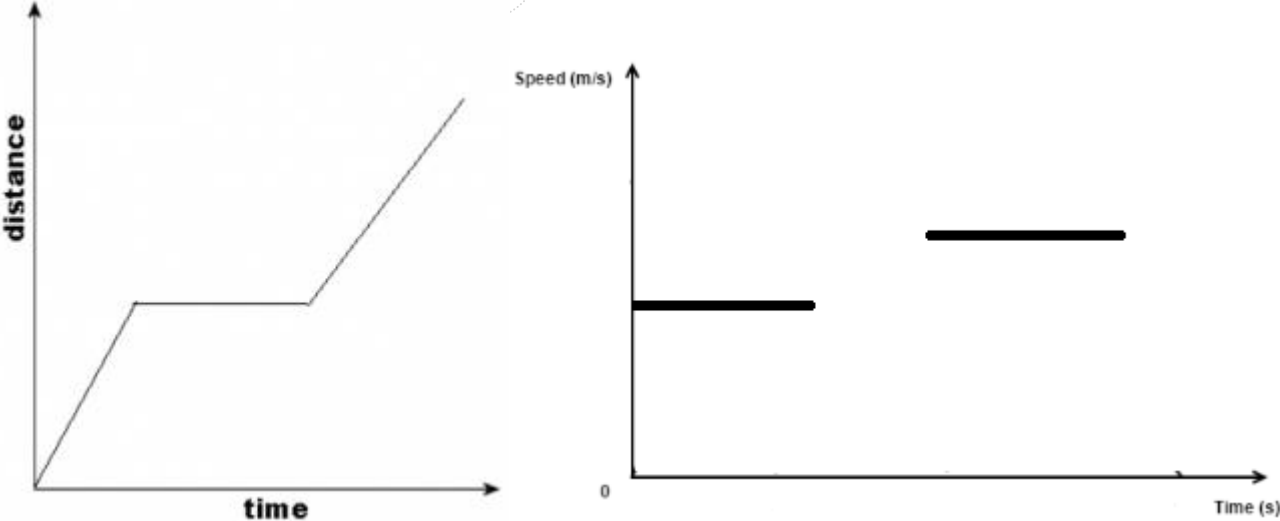
Topic 22: Graphs in practical situations (Kinematics)



A Velocity - Time Graph

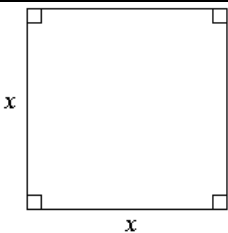
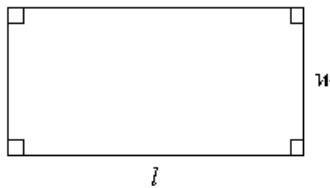
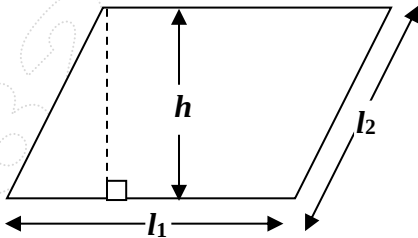
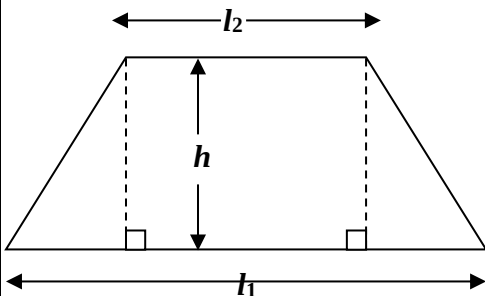
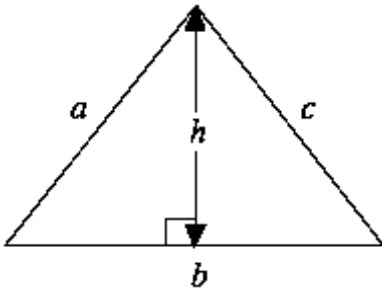
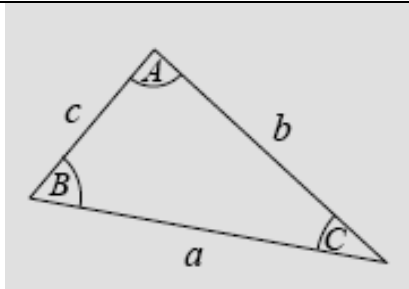


The distance travelled is area under graph.
The acceleration and deceleration can be found by
finding the gradient of the lines.



Topic 23: Mensuration

Most Common Formulae Use in Mensuration

<u>2D Shapes</u>	
<u>Square</u> Area of the square = x^2 Perimeter of the square = $4x$	
<u>Rectangular</u> Area of the rectangular = $l \times w$ Perimeter of the rectangular = $2(l + w)$	
<u>Parallelogram</u> Area of the Parallelogram = (Base $\times$ Height) Perimeter of the Parallelogram = 2 (Sum of two adjacent sides)	
<u>Trapezium</u> Area of the trapezium = $\frac{1}{2} \times h (l_1 + l_2)$	
<u>Triangle</u> Area of the triangle = $\frac{1}{2} h \times b$ Perimeter of the triangle = Sum of all sides = $a + b + c$	
There are two useful formulae for the area of a triangle when (i) the length of two sides and the included angle are known. Area = $\frac{1}{2} ab \sin C = \frac{1}{2} ac \sin B = \frac{1}{2} bc \sin A$ (ii) the lengths of three sides are known.	

Circle

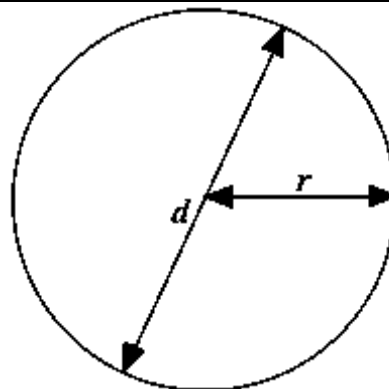
The *circumference* of a circle can be calculated using

$$C = 2\pi r \quad \text{or} \quad C = \pi d$$

Where r is the radius and d the diameter of the circle.

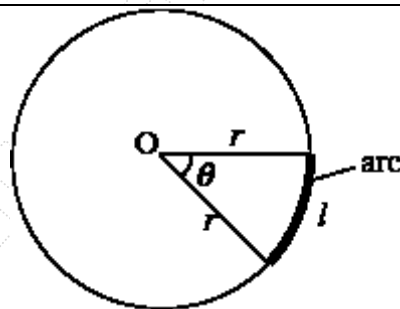
The area of a circle is found using

$$A = \pi r^2 \quad \text{or} \quad A = \frac{\pi D^2}{4}$$



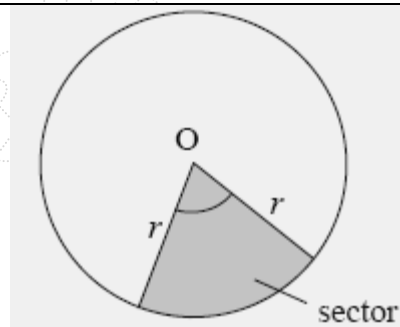
A part of the circumference of a circle is called an arc. If the angle subtended by the arc at the centre of the circle is θ then the arc length l is given by

$$l = \frac{\theta}{360^\circ} \times 2\pi r$$



The region between the two radii and the arc is called a sector of the circle. The area of the sector of the circle is $\frac{\theta}{360^\circ} \times \pi r^2$

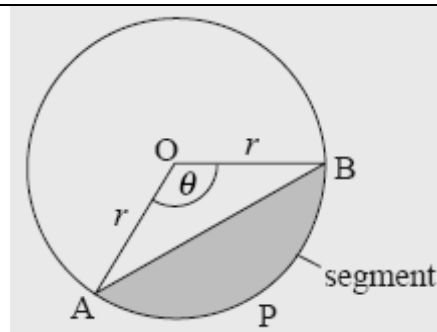
$$A = \frac{\theta}{360^\circ} \times \pi r^2$$



The region between the chord AB and the arc APB is called a **segment of the circle**. The area of the segment of the circle is

$$\text{Area of minor Segment} = \frac{\theta}{360^\circ} \times \pi r^2 - \frac{1}{2} r^2 \sin \theta$$

$$\text{Area of major Segment} = \frac{360^\circ - \theta}{360^\circ} \times \pi r^2 + \frac{1}{2} r^2 \sin \theta$$

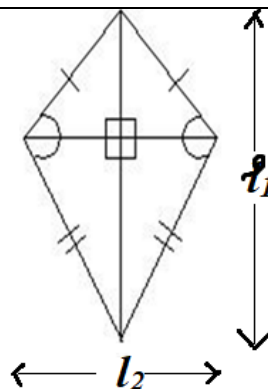


Kite

Area of Kite = $\frac{1}{2}$ (Sum of Length of its Diagonals)

Area of the Kite = $\frac{1}{2} \times (l_1 + l_2)$

Perimeter of the Kite = 2 (Sum of two unequal sides)



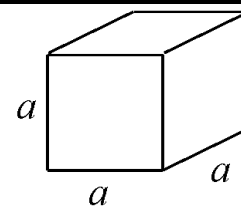
3D Shapes

Cube

Volume of the Cube = a^3

Total Surface area of the Cube = $6 a^2$

Surface area of the open Cube/box (without top) = $5 a^2$



Cuboid

Volume of the Cuboid = $L \times B \times H$

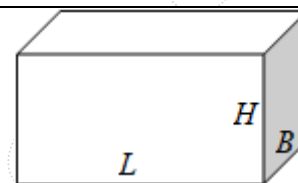
Total Surface area of the Cuboid = $2 (L \times B + B \times H + L \times H)$

Surface area of the open Cuboid /box (without top)

= $L \times B + 2 (B \times H + L \times H)$

Surface area four walls of the Cuboid /box

(without top and base) = $2(L \times H + B \times H)$



Cylinder

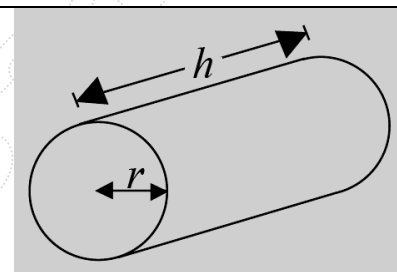
Volume of the cylinder = $\pi r^2 h$

Total Surface area of the cylinder = $2 \pi r (r + h)$

or = $2 \pi r^2 + 2 \pi r h$

Surface area of the without led/top(Open Cylinder) = $\pi r (r + 2h)$

Surface area of the Both side open Cylinder(Hollow pipe) = $2 \pi r h$



Prism

Volume of the prism = $A l$

Total Surface area of the prism = **Sum of area of all faces**

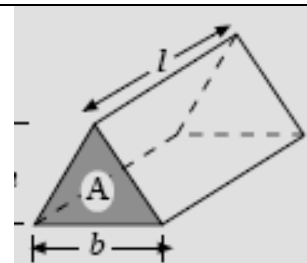
Volume of Rectangular Prism: $L \times B \times \text{Figure Height}$

Volume of Triangular Prism: $\frac{1}{2} \times B \times H \times \text{Figure Height}$

Volume of Trapezium Prism:

= $\frac{1}{2} \times H \times (\text{Sum of 2 Parallel sides}) \times \text{Figure Height}$

Volume of Prism Cuboids: **Base Area** $\times$ **Figure Height**



Formulas Given in Paper

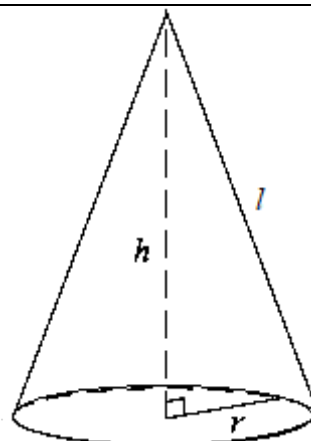
Cone

Volume of the cone = $\frac{1}{3} \pi r^2 h$

$$r^2 + h^2 = l^2$$

Total Surface area of the cone = $\pi r^2 + \pi r l = \pi r (r + l)$

Curve surface area of Cone = $\pi r l$



Sphere

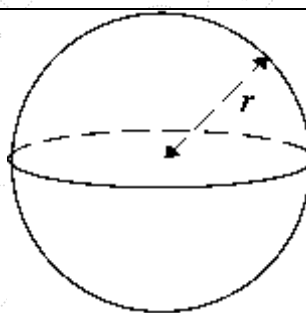
Volume of the sphere = $\frac{4}{3} \pi r^3$

Volume of the hemi sphere = $\frac{2}{3} \pi r^3$

Curve Surface area of the sphere = $4\pi r^2$

Curve Surface area of the hemi sphere = $2\pi r^2$

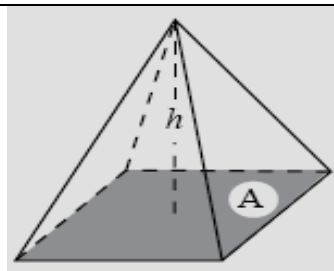
Total Surface area of the hemi sphere (with top) = $3\pi r^2$



Pyramid

Volume of the Pyramid = $\frac{1}{3} (\text{Base Area} \times \text{Height})$

Curve Surface area of the Pyramid = **Sum of area of all faces**



Topic 24: Similarity and congruence

Topic 24(a): Congruent Figures

Identical and same in size figure are congruent

Properties

If $\triangle ABC \cong \triangle XYZ$ or $\triangle ABC \equiv \triangle XYZ$

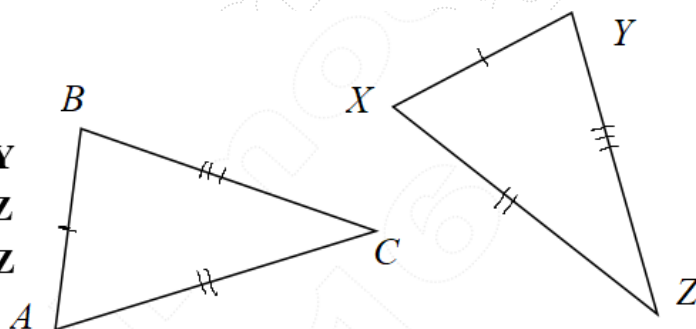
Then $\hat{A} = \hat{X}$ and $AB = XY$
 $\hat{B} = \hat{Y}$ and $BC = YZ$
 $\hat{C} = \hat{Z}$ and $AC = XZ$
 The corresponding sides and angles are equal.

Conditions for Congruency:

1- Side-side-side(S.S.S)

All sides of one triangle equal to the corresponding sides of the other triangle.

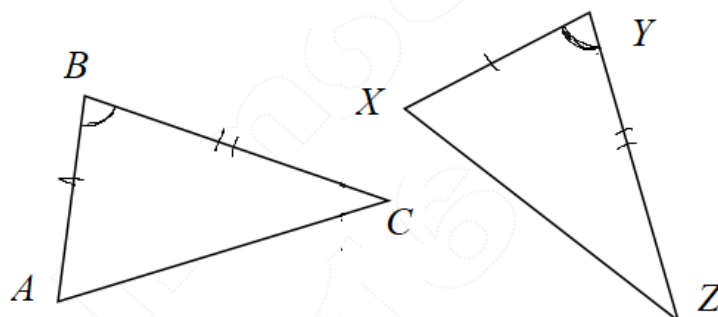
$$\begin{aligned} AB &= XY \\ BC &= YZ \\ AC &= XZ \end{aligned}$$



2- Side-Angle-Side (S.A.S)

Two sides and included angle of one triangle equal to the corresponding sides and included angle of the other triangle.

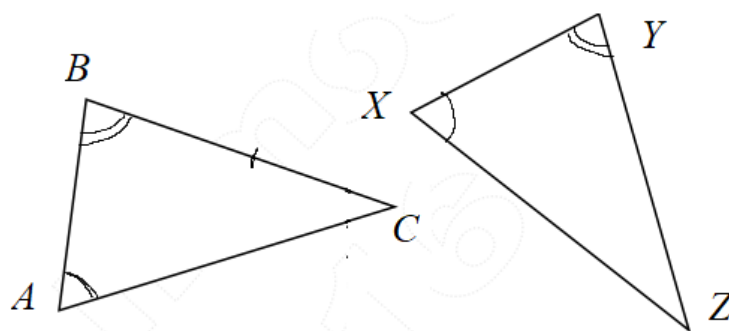
$$\begin{aligned} AB &= XY \\ \hat{B} &= \hat{Y} \\ BC &= YZ \end{aligned}$$



3- Angle-angle-side (A.A.S)

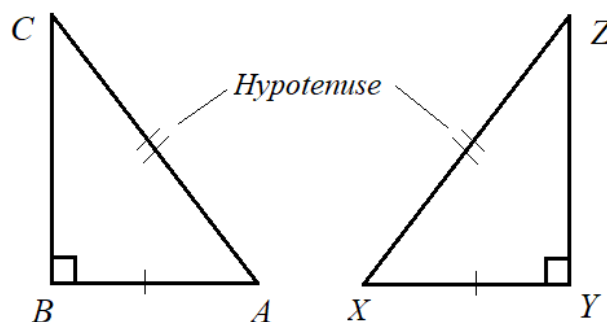
Two angles and one side of one triangle equal to the corresponding two angles and one side of the other triangle.

$$\begin{aligned} \hat{A} &= \hat{X} \\ \hat{B} &= \hat{Y} \\ BC &= YZ \end{aligned}$$



4- If $\triangle ABC \cong \triangle XYZ$ or $\triangle ABC \equiv \triangle XYZ$

$\hat{B} = \hat{Y} = 90^\circ$ and
 $AC = XY$ (Hypotenuse)
 $AB = XY$ (side)



Topic 24(b): Similar Figures

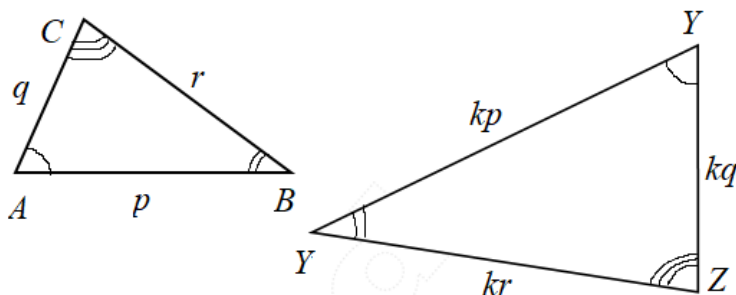
The figures are said to be similar if they are same shape-wise.

Properties

If $\triangle ABC \sim \triangle XYZ$

Then $\hat{A} = \hat{X}$, $\hat{B} = \hat{Y}$, $\hat{C} = \hat{Z}$ and

$$\frac{AB}{XY} = \frac{BC}{YZ} = \frac{AC}{XZ} \quad \left(= \frac{1}{k}\right)$$

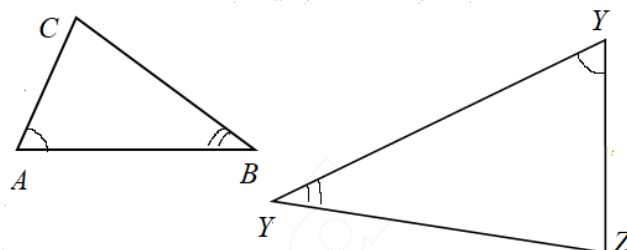


The corresponding angles are equal and corresponding sides are proportional.

Conditions for Similarity:

1-Two pairs of angles are congruent

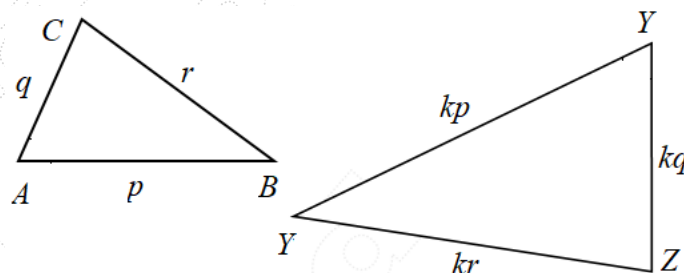
$$\text{If } \hat{A} = \hat{X}, \hat{B} = \hat{Y}, \hat{C} = \hat{Z}$$



2-The Corresponding sides are proportional.

$$\frac{AB}{XY} = \frac{BC}{YZ} = \frac{AC}{XZ} \quad \left(= \frac{1}{k}\right)$$

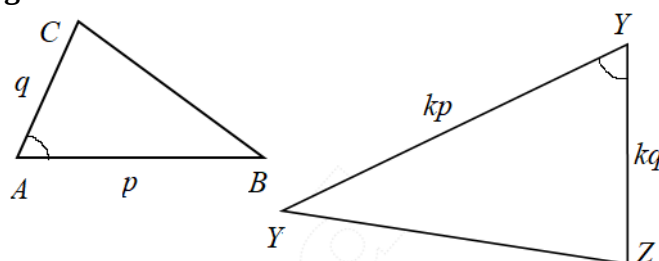
$$\frac{p}{kp} = \frac{q}{kq} = \frac{r}{kr} = \frac{1}{k}$$



3-Two sides are proportional and the included angle is same.

$$\frac{AB}{XY} = \frac{BC}{YZ} = \frac{1}{k} \quad \text{and} \quad \hat{B} = \hat{Y},$$

$$\frac{p}{kp} = \frac{q}{kq} = \frac{1}{k}$$



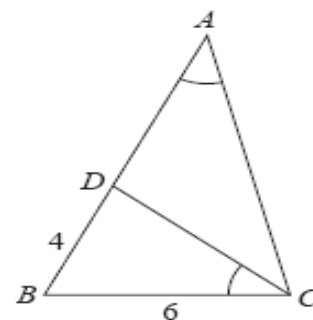
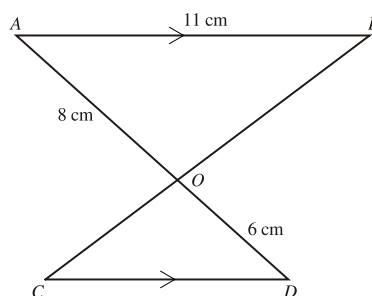
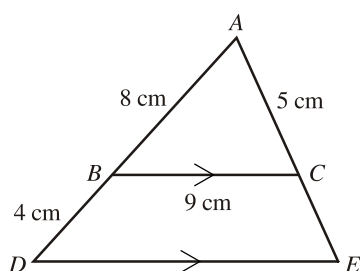
Important Result

Ratio of perimeter of similar figures = Ratio of corresponding sides

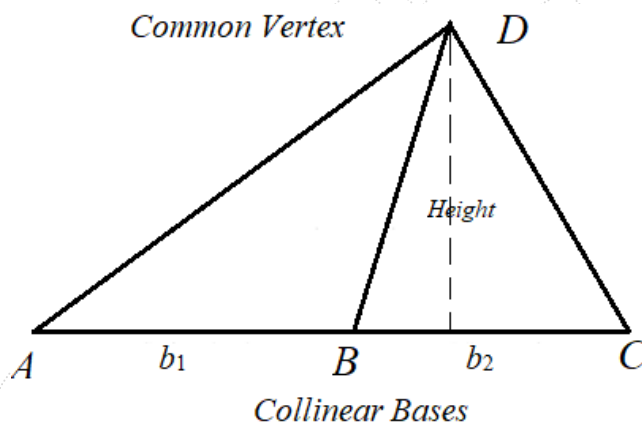
Ratio of Area (Surface Area) of similar figures = (Ratio of corresponding sides)²

Ratio of Volume or Mass of similar figures = (Ratio of corresponding sides)³

Some Examples of Similar Triangles

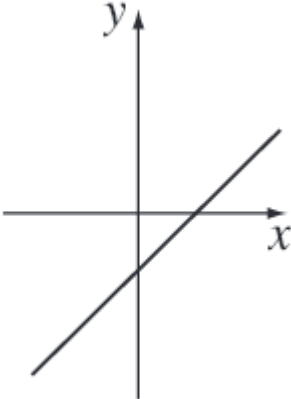
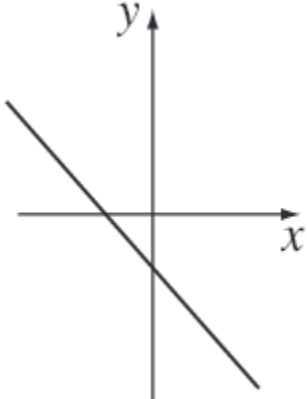
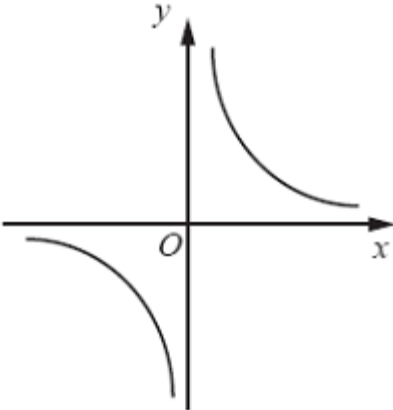
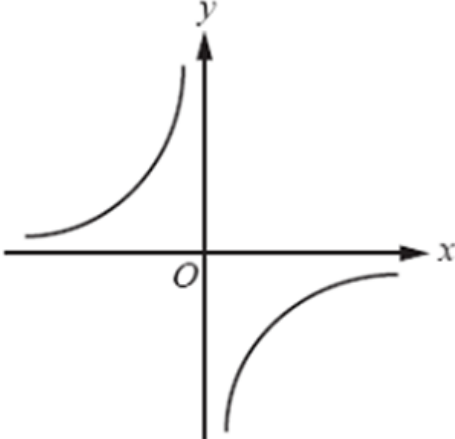
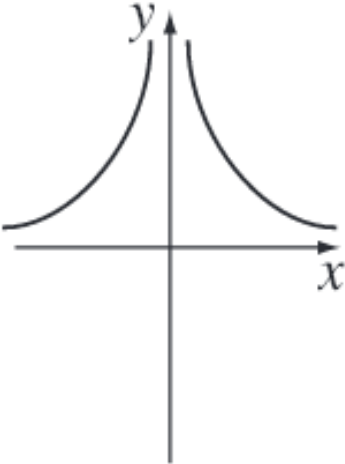
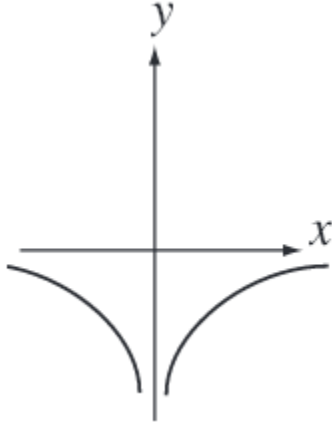


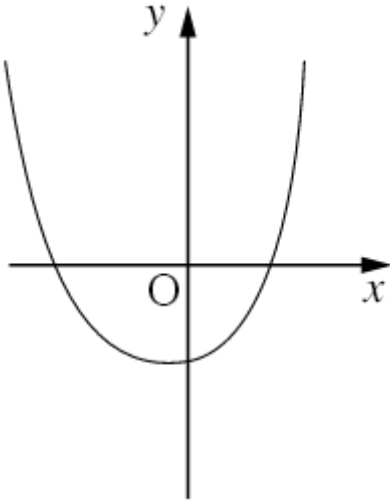
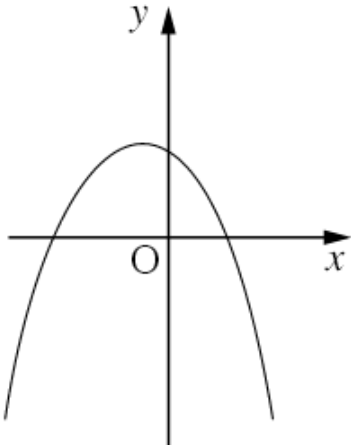
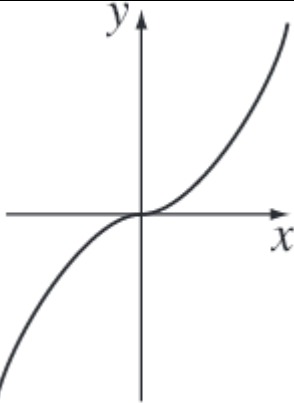
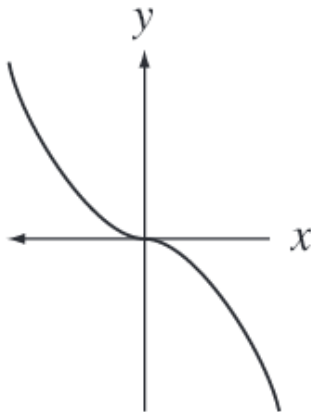
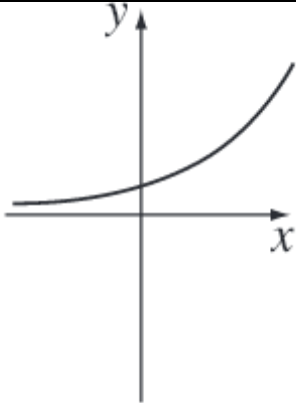
- Ratio of Diameter/Height/Radius/Circumference/Perimeter of similar figures = Ratio of corresponding sides or $(l_1 : l_2)$
- Ratio of Area of similar figures = (Ratio of corresponding sides)² or $(l_1 : l_2)^2$
- Ratio of Volume/Masses of similar figures = (Ratio of corresponding sides)³ or $(l_1 : l_2)^3$
- Ratio of Area of Triangles having collinear bases and common vertex = (Ratio of their bases)
Area of triangle ABD : Area of triangle BCD = $b_1 : b_2$



Topic 25: Graphs of Functions

Sketch of Some Important Graphs

	$y = mx + c$, $m > 0$ & $c < 0$	$y = mx + c$, $m < 0$ & $c < 0$
Linear Equation		
Reciprocal Equation	$y = \frac{a}{x}$, $a > 0$,	$y = \frac{a}{x}$, $a < 0$
		
Reciprocal Equation	$y = \frac{a}{x^2}$, $a > 0$,	$y = \frac{a}{x^2}$, $a < 0$
		

Quadratic Equation	$y = ax^2 + bx + c$, $a > 0$	$y = ax^2 + bx + c$, $a < 0$
		
Cubic Equation	$y = ax^3$, $a > 0$	$y = ax^3$, $a < 0$
		
	$y = a^x$, $a > 0$	
Exponential Equation		

Topic 26: Geometrical constructions, Loci

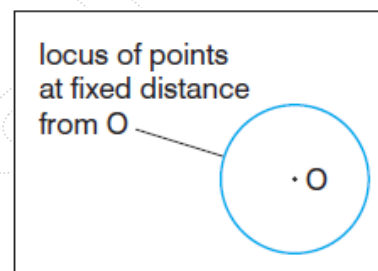
Geometrical constructions (Loci)

A point moving so that it is a fixed distance from a point O

A set of points or Locus points 5 cm from A. (A circle with centre A and radius 5 cm).

A set of points or region more than 5 cm from A. (Outside the circle)

A set of points or region Less than 5 cm from A. (Inside the circle)

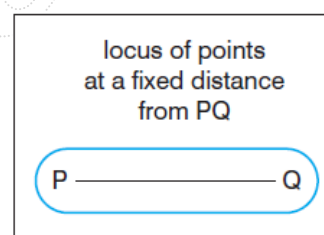


A point moving so that it is a fixed distance from fixed line. (Parallel Line)

A set of points or Locus points 5 cm from AB. (A circle with centre A and radius 5 cm).

A set of points or region more than 5 cm from AB. (Outside the circle)

A set of points or region Less than 5 cm from AB. (Inside the circle)

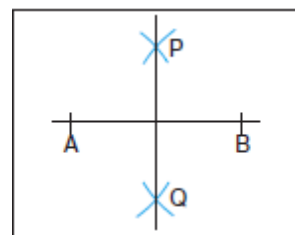


A point moving so that it is a equidistance from two fixed points A and B (Perpendicular Bisector)

A set of points or Locus points equidistance from A & B. (Perpendicular Bisector of AB).

A set of points or Locus points nearer to A than B.

A set of points or Locus points nearer to B than A.

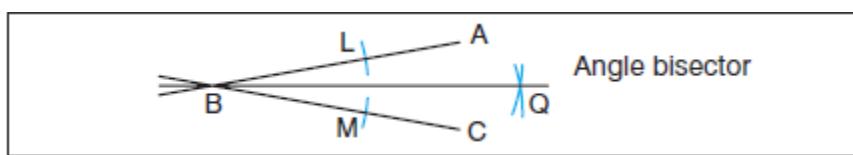


A point moving so that it is a equidistance from two fixed points AB and BC .(Angle Bisector)

A set of points or Locus points equidistance from AB & BC. (Angle Bisector of angle ABC).

A set of points or Locus points nearer to AB than BC.

A set of points or Locus points nearer to BC than AB.



Topic 27: Categorical, numerical and grouped data

Measures of central tendency (Averages)

Mode

The mode of a distribution is the value that occurs most frequently.

e.g. Find the mode of the following distribution.

8, 6, 7, 4, 9, 8, 8, 6, 7, 6, 8

The number 8 occurs most frequently so the mode is 8.

Median

The median of a distribution is the middle value when the values are arranged in numerical order.

e.g. Find the median of the following distribution.

8, 6, 7, 4, 9, 8, 8, 6, 7, 6, 8

Rearranging in numerical order:

4, 6, 6, 6, 7, 7, 8, 8, 8, 8, 9

So the median = 7

e.g. Find the median of the following distribution.

8, 6, 7, 4, 9, 8, 8, 6, 7, 6, 8, 10

Rearranging in order:

4, 6, 6, 6, 7, 7, 8, 8, 8, 8, 9, 10

So the median = $(7 + 8) / 2 = 7.5$

Mean

The mean (or arithmetic mean) of a distribution is found by summing the values of the distribution and dividing by the number of values.

e.g. Find the mean of the following distribution.

8, 6, 7, 4, 9, 8, 8, 6, 7, 6, 8

Mean = $77 / 11 = 7$

Mean of a frequency distribution

$$\text{Mean} = \frac{\text{sum of values}}{\text{number of value}} \text{ or } \text{Mean} = \frac{\Sigma \text{frequency} \times \text{values}}{\Sigma \text{frequencies}} = \frac{\Sigma fx}{\Sigma f}$$

Range

The **range** of a distribution is found as the numerical difference between the greatest value and least value.

The range should always be given as a *single* value.

e.g. Find the range of the following test marks.

9 7 8 10 9 8 8 2 9 10 8

Range = greatest value – least value

= $10 - 2 = 8$

Topic 28: Statistical Diagrams

Cumulative frequency diagrams

A cumulative frequency diagram (or **ogive**) can be used to find the median and quartiles of a distribution. To find the cumulative frequency, find the accumulated totals and plot them against the data values. The cumulative frequency diagram is formed by joining the points with a smooth curve.

Using the cumulative frequency (2)

The cumulative frequency diagram can be used to find out other information about the distribution such as how many customers waited:

- less than 5 minutes
- more than 10 minutes.

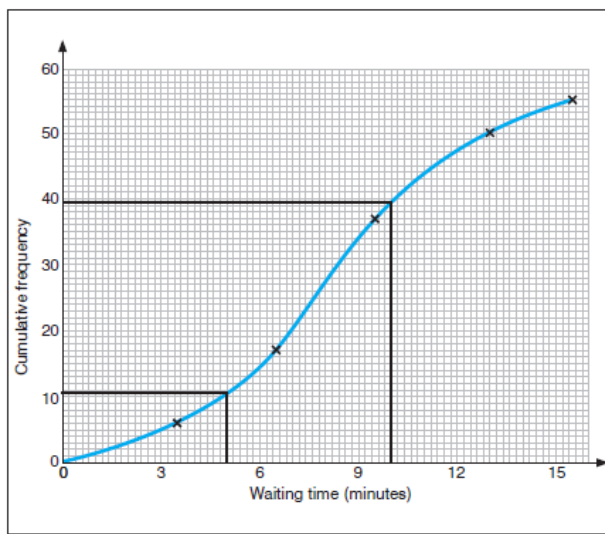
Waiting time (minutes)	Frequency	Cumulative frequency
1-3	6	6
4-6	11	17
7-9	20	37
10-12	13	50
13-15	5	55

- To find out how many customers waited less than 5 minutes, read the cumulative frequency for a waiting time of 5 minutes. From the graph, the number of customers who waited less than 5 minutes is 10.5. An answer rounded to 10 or 11 would be acceptable.
- To find out how many customers waited more than 10 minutes, read the cumulative frequency for a waiting time of 10 minutes.

From the graph, the number of customers who waited less than 10 minutes is 39.5.

So the number of customers who waited more than 10 minutes is $55 - 39.5 = 15.5$, or 16 customers to the nearest whole number.

NB This information cannot just be read from the table as the 10-12 group contains times from 9.5 minutes to 12.5 minutes.



Using the cumulative frequency (1)

The cumulative frequency diagram can be used to find the median and quartiles of the given distribution.

Median

You have already seen that the median is the middle value and, in general, the median is the $\frac{1}{2}(n + 1)$ th value in the distribution (see Handling data card 3).

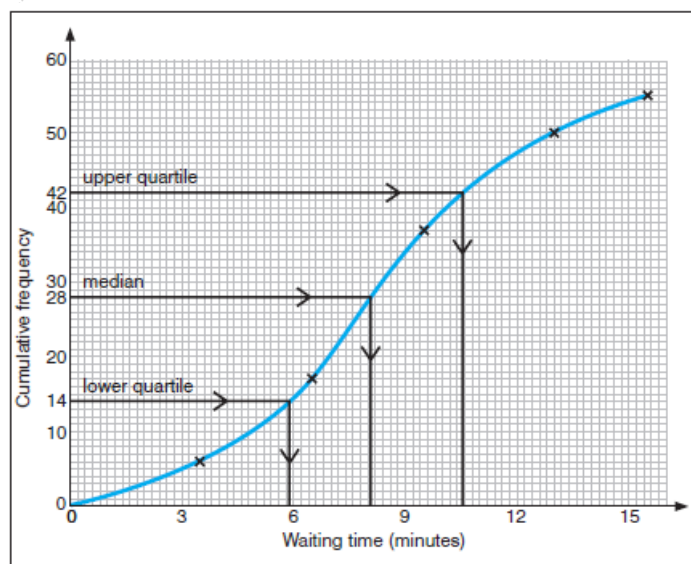
For the cumulative frequency diagram on Handling data card 6, the median is given by the $\frac{1}{2}(n + 1)$ th value $= \frac{1}{2}(55 + 1) = 28$ th value in the distribution.

Interquartile range

The interquartile range is the difference between the upper quartile and the lower quartile where the lower quartile is the $\frac{1}{4}(n + 1)$ th value and the upper quartile is the $\frac{3}{4}(n + 1)$ th value in the distribution (see Handling data card 5).

For the cumulative frequency diagram the lower quartile is given by the $\frac{1}{4}(n + 1) = \frac{1}{4}(55 + 1) = 14$ th value and the upper quartile is given by the $\frac{3}{4}(n + 1) = \frac{3}{4}(55 + 1) = 42$ nd value.

NB Your answers should usually be correct to the nearest half-square (in the example on the right, ± 0.1 minutes).



From the graph:

median = 8.1

upper quartile = 10.6

lower quartile = 5.9

interquartile range

$$= \text{upper quartile} - \text{lower quartile} = 10.6 - 5.9 = 4.7$$

Histogram – unequal class intervals

Histograms are like bar charts except that it is the area of each 'bar' that represents the frequency.

The bars of the histogram should be drawn at the class boundaries and the area of the bars should be proportional to the frequency,

i.e. $\text{class width} \times \text{height} = \text{frequency}$ so that
 $\text{height} = \text{frequency} \div \text{class width}$.

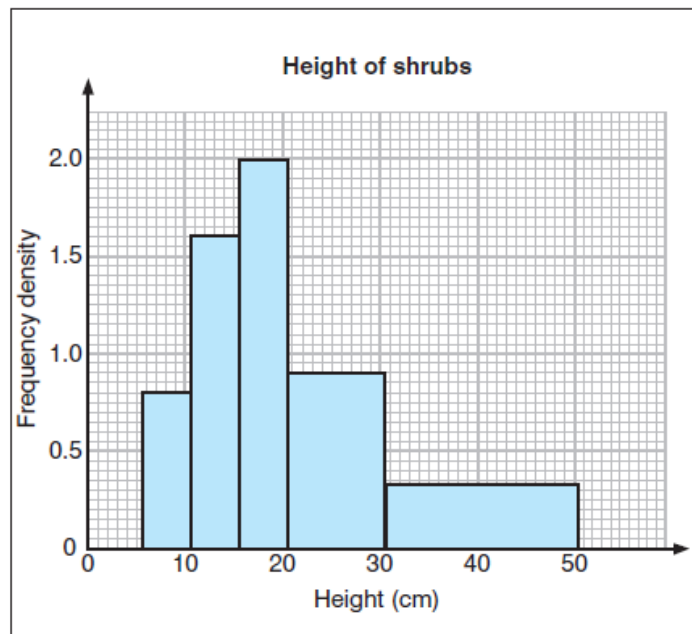
The height is referred to as the frequency density so:
frequency density = frequency $\div$ class width

e.g. The following frequency distribution shows the height of 38 shrubs measured to the nearest centimetre. Draw the histogram.

Height (cm)	5–9	10–14	15–19	20–29	30–50
Frequency	4	8	10	9	7

To draw a histogram you need to calculate the frequency densities as follows.

Height	Frequency	Class width	Frequency density
5–9	4	5	$4 \div 5 = 0.8$
10–14	8	5	$8 \div 5 = 1.6$
15–19	10	5	$10 \div 5 = 2.0$
20–29	9	10	$9 \div 10 = 0.9$
30–50	7	21	$7 \div 21 = 0.3333...$



NB Since the height is measured to the nearest centimetre then the 5–9 interval extends from 4.5 to 9.5, giving it a width of 5 cm, etc.

Frequency polygons

A frequency polygon can be drawn from a bar chart or histogram by joining the midpoints of the tops of the bars, in order, with straight lines to form a polygon.

The lines should be extended to the horizontal axis so that the area under the frequency polygon is the same as the area under the bar chart or histogram.

e.g. Draw the histogram for the following information.

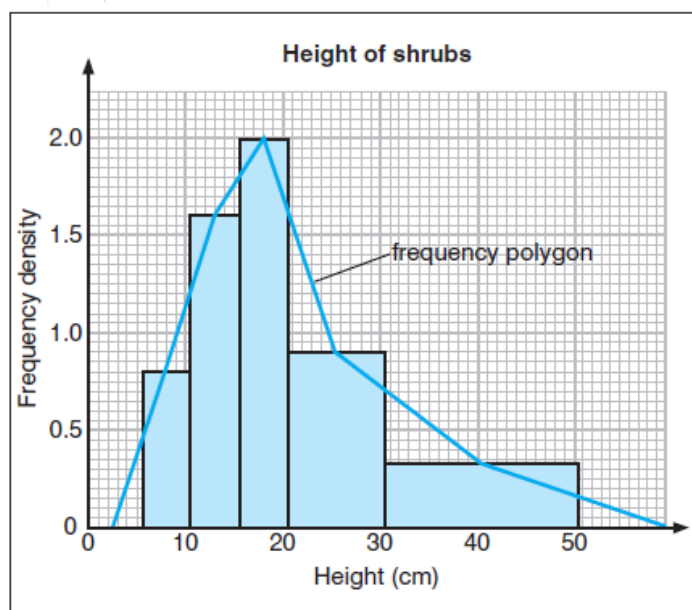
Height (cm)	5–9	10–14	15–19	20–29	30–50
Frequency	4	8	10	9	7

To draw the frequency polygon, calculate the frequency densities as before.

Height	Frequency	Class width	Frequency density
5–9	4	5	$4 \div 5 = 0.8$
10–14	8	5	$8 \div 5 = 1.6$
15–19	10	5	$10 \div 5 = 2.0$
20–29	9	10	$9 \div 10 = 0.9$
30–50	7	21	$7 \div 21 = 0.3333...$

The frequency polygon is then plotted at the mid-interval values i.e. 7, 12, 17, 24.5 and 40.

NB It is often helpful to draw the frequency polygon over the top of the completed histogram.



NB The lines are extended to the horizontal axis on each side so that the area under the frequency polygon is the same as the area under the histogram (see Handling data card 15).

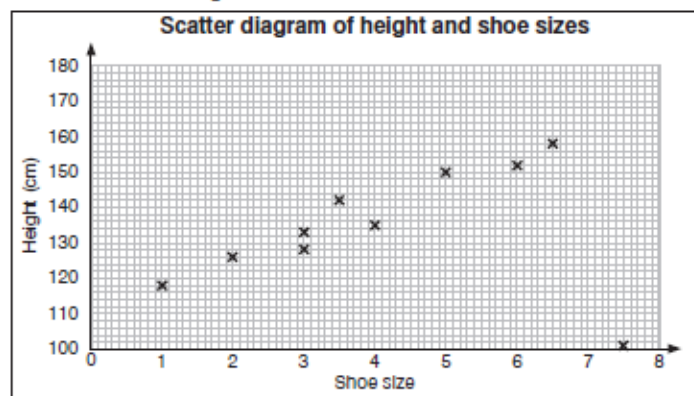
Scatter diagrams

Scatter diagrams (or **scatter graphs**) are used to show the relationship between two variables. Each of the two variables is assigned to a different axis and the information is plotted as a series of coordinates on the scatter diagram.

e.g. Draw a scatter graph of the following table which shows the heights and shoe sizes of 10 pupils.

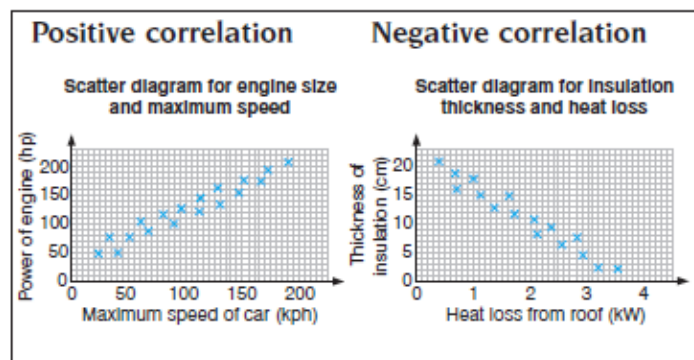
Shoe size	3	2	$6\frac{1}{2}$	4	3	6	1	$3\frac{1}{2}$	$7\frac{1}{2}$	5
Height (cm)	133	126	158	135	128	152	118	142	101	150

To draw the scatter graph, consider each pair of values as a different pair of coordinates (3, 133), (2, 126), ($6\frac{1}{2}$, 158), etc.



Positive and negative correlation

Correlation can also be described in terms of positive and negative correlation.



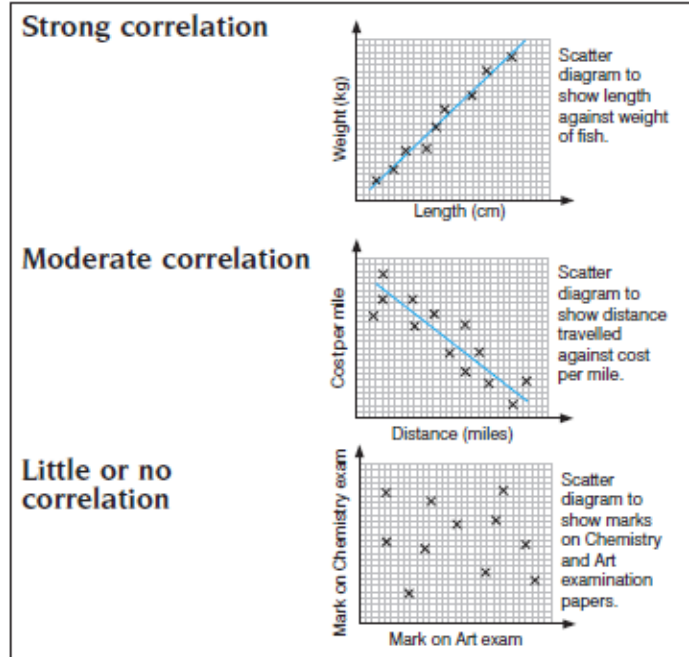
Where an increase in one variable is associated with an increase in the other variable, the correlation is **positive** or **direct**; where an increase in one variable is associated with a decrease in the other variable, the correlation is **negative** or **inverse**.

Line of best fit

When the points on a scatter diagram show moderate or strong correlation, a straight line can be drawn through, or as close to, as many of the points as possible, to approximate the relationship. This is the

Correlation

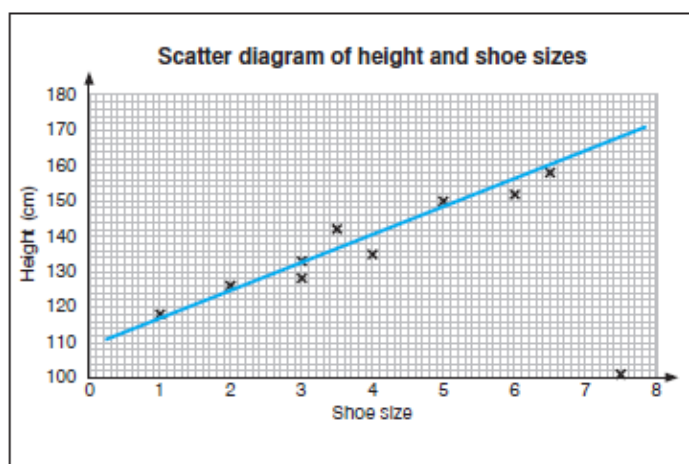
Scatter graphs can be used to show whether there is any relationship or **correlation** between the two variables. The following descriptions of such relationships or correlation need to be known.



line of best fit (or **regression line**). It can be used to predict other values from the given data.

e.g. Draw a scatter graph and line of best fit for the following table which shows the heights and shoe sizes of 10 pupils.

Shoe size	3	2	$6\frac{1}{2}$	4	3	6	1	$3\frac{1}{2}$	$7\frac{1}{2}$	5
Height (cm)	133	126	158	135	128	152	118	142	101	150



NB In most cases a line of best fit can be drawn 'by eye'. For more accurate work the line of best fit should pass through $(\bar{x}, \bar{y})$ where $\bar{x}$ and $\bar{y}$ are the mean values of x and y respectively.

Topic 29: Probability

Probability

In probability, an event which is certain to happen will have a probability of 1 whereas an event which is impossible will have a probability of 0. In general,

$$p(\text{success}) = \frac{\text{number of 'successful' outcomes}}{\text{number of possible outcomes}}$$

where $p(\text{success})$ means 'the probability of success'.

e.g. If a box contains seven red counters, ten blue counters and eight yellow counters then:

- the probability of picking a red counter is:

$$p(\text{red counter}) = \frac{\text{number of red counters}}{\text{number of counters}} = \frac{7}{25}$$

- the probability of picking a blue counter is:

$$p(\text{blue counter}) = \frac{\text{number of blue counters}}{\text{number of counters}} = \frac{10}{25} = \frac{2}{5}$$

Cancelling down where possible.

- the probability of picking a green counter is:

$$p(\text{green counter}) = \frac{\text{number of green counters}}{\text{number of counters}} = \frac{0}{25} = 0$$

i.e. the outcome is impossible.

Possibility spaces

A possibility space is a diagram which can be used to show the outcomes of various events.

e.g. If two fair dice are thrown and the sum of the scores is noted, then the following diagram can be drawn to illustrate the possible outcomes.

		Second die					
		1	2	3	4	5	6
First die	1	2	3	4	5	6	7
	2	3	4	5	6	7	8
	3	4	5	6	7	8	9
	4	5	6	7	8	9	10
	5	6	7	8	9	10	11
	6	7	8	9	10	11	12

The diagram above is a possibility space; there are 36 possible outcomes.

From the diagram:

- the probability of a sum of 2 is $\frac{1}{36}$
(as there is only one entry with a sum of 2)
 - the probability of a sum of 7 is $\frac{6}{36} = \frac{1}{6}$
(as there are six entries with a sum of 7)
 - the probability of a sum of 10 is $\frac{3}{36} = \frac{1}{12}$
(as there are three entries with a sum of 10)
- etc.

Total probability

The probability of an event happening is equal to 1 *minus* the probability of the event not happening.

e.g. If the probability that it will rain tomorrow is $\frac{1}{5}$ then the probability that it will not rain tomorrow is $1 - \frac{1}{5} = \frac{4}{5}$.

$$\text{i.e. } p(\text{not rain}) = 1 - p(\text{rain}) = 1 - \frac{1}{5} = \frac{4}{5}$$

Theoretical and experimental

Theoretical probability is based on equally likely outcomes and is used to predict how an event should perform in theory; **experimental probability** (or **relative frequency**) is used to predict how an event performs in an experiment.

e.g. The following frequency distribution is obtained when a die is thrown 100 times.

Score	1	2	3	4	5	6
Frequency	15	19	18	15	17	16

The theoretical probability of a score of six is $\frac{1}{6}$ and the experimental probability (or relative frequency) of a score of six is $\frac{16}{100} = \frac{4}{25}$.

NB The expected number of sixes when a die is thrown 100 times is:

$$100 \times \frac{1}{6} = 16.\dot{6}$$

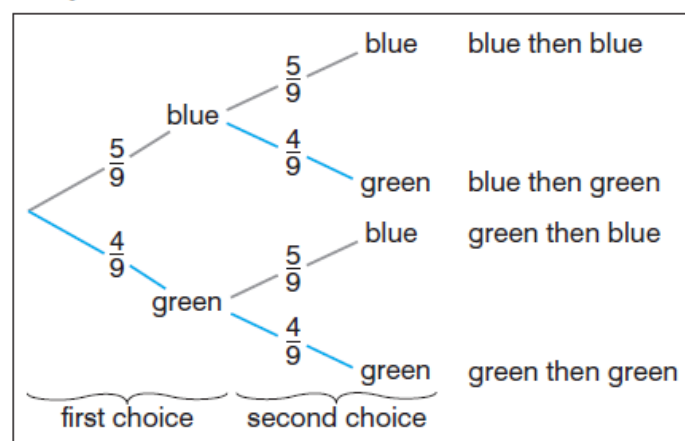
or 17 (to the nearest whole number).

Tree diagrams

A tree diagram is a diagram where the probabilities of different events are written on different branches, so that the probabilities on individual pairs (or groups) of branches always add up to one.

e.g. A bag contains five blue and four green discs. A disc is drawn from the bag, replaced and then a second disc is drawn from the bag. Draw a tree diagram to show the various possibilities that can occur.

The following tree diagram shows the various possibilities that can occur.

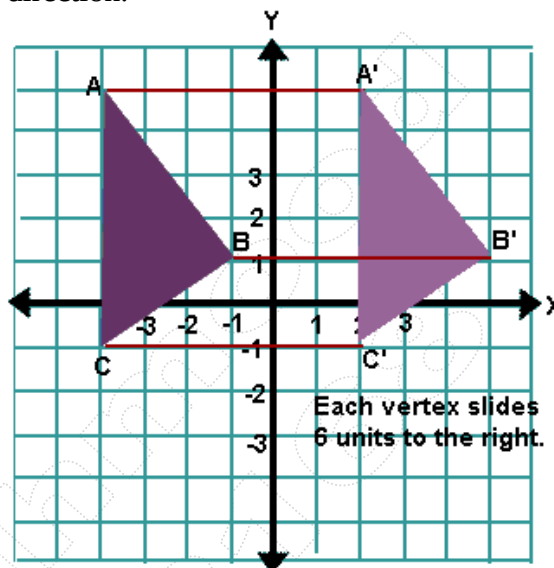
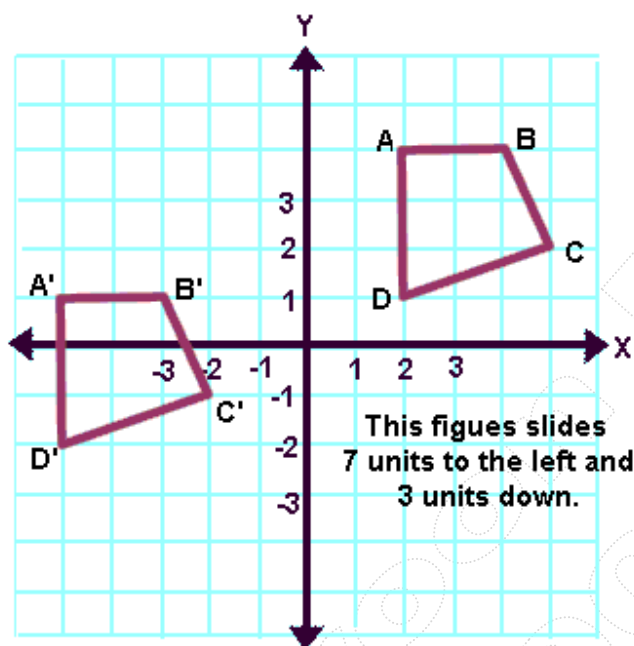


Topic 30: Transformation

Translation

A translation "slides" an object a fixed distance in a given direction. The original object and its translation have the same shape and size, and they face in the same direction.

The example shows how each vertex moves the same distance in the same direction.



In this example, the "slide" moves the figure 7 units to the left and 3 units down. (or 3 units down and 7 units to the left.)

	Object	Translation Vector	Image
CASE -I	✓	✓	?

$$\text{Translation Vector} = \begin{pmatrix} \text{Horizontal Movement} \\ \text{Vertical Movement} \end{pmatrix}$$

Move object horizontally (Rightward movement for +ve and Leftward movement for – ve) and vertically (Upward movement for +ve and Downward movement for – ve) according

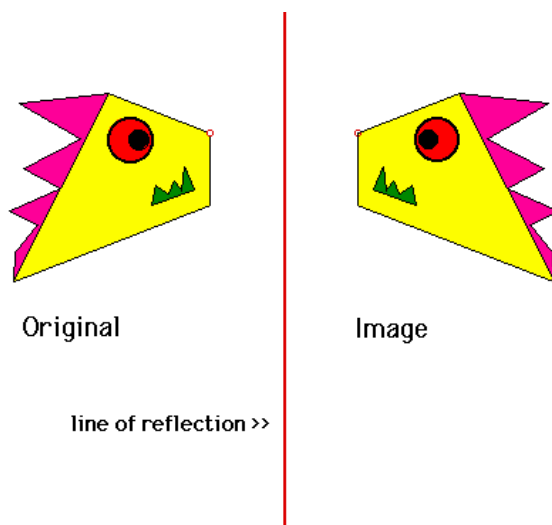
	Object	Translation Vector	Image
CASE -II	✓	?	✓

Those transformation in which **objects** are **Slide or Shift** to form **Image** is Translation.

To find **Translation Vector** start moving from any point of **object** to corresponding point of **Image** and check horizontal and vertical movement to find translation vector.

Reflection

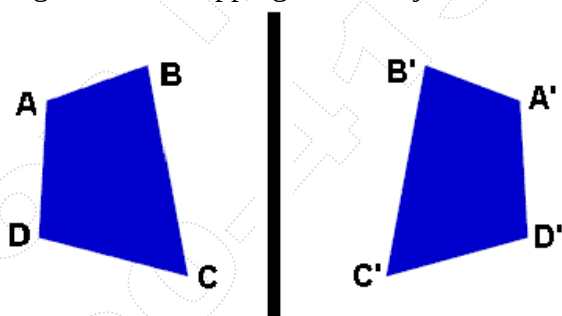
A **reflection** can be seen in water, in a mirror, in glass, or in a shiny surface. An object and its reflection have the **same shape and size**, but the **figures face in opposite directions**. In a mirror, for example, right and left are **interchanged**.



Line reflections are FLIPS!!!

The line (where a mirror may be placed) is called the **line of reflection**. The distance from a point to the line of reflection is the same as the distance from the point's image to the line of reflection.

A reflection can be thought of as a "flipping" of an object over the line of reflection.



If you folded the two shapes together **line of reflection** the two shapes would overlap exactly!

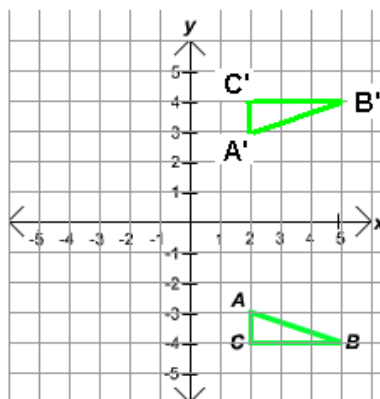
	Object	Mirror Line	Image
CASE -I	✓	✓	?
CASE -II	✓	?	✓

Describe Single Transformation that map A onto B

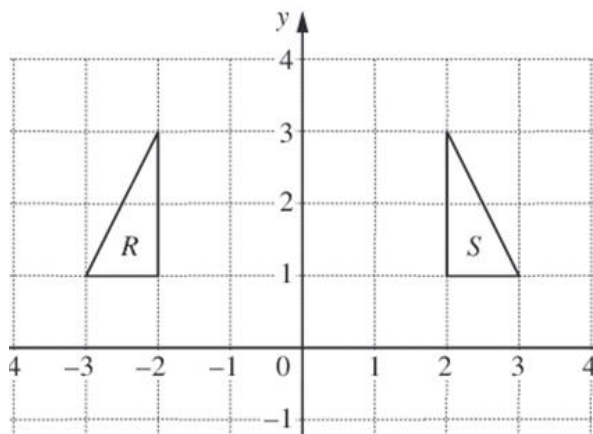
STANDARD REFLECTIONS

Reflection About x-axis

Every (x, y) change into $(x, -y)$

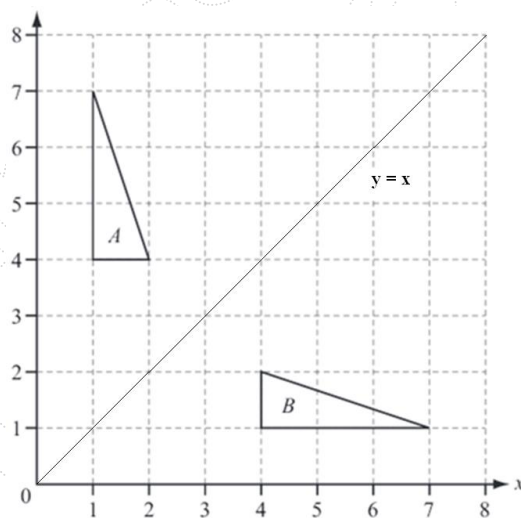


Every (x, y) change into $(-x, y)$



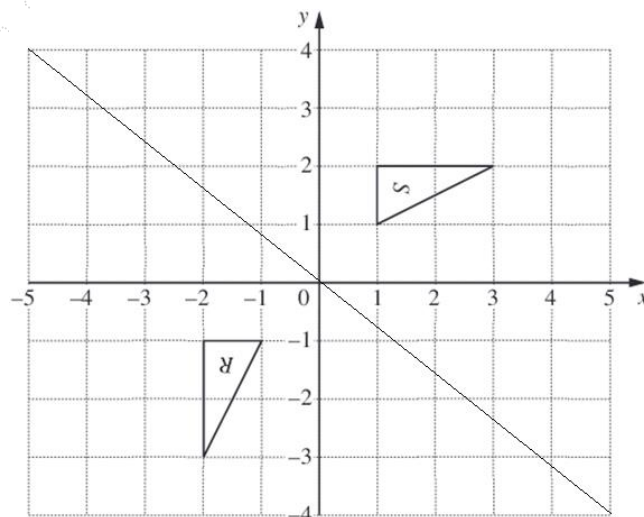
Reflection About $y = x$

Every (x, y) change into (y, x)



Reflection About $y = -x$

Every (x, y) change into $(-y, -x)$



Reflection About $y = mx + c$

Measure perpendicular distance each point of object and mark on opposite side of **Mirror Line** at same distance.

	Object	Mirror Line	Image
CASE -II	✓	?	✓

Describe Single Transformation that map A onto B

Those transformation in **object** are **FLIP** to form **Image** are Reflection.

To find Mirror line draw, **perpendicular bisector** of any two corresponding points of **Image** and **Object**.

Rotation

A **rotation** is a transformation that turns a figure about a fixed point called the center of rotation. An object and its rotation are the **same shape and size**, but the **figures may be turned in different directions**.

Rotations are **TURNS!!!**

Angle of Rotation

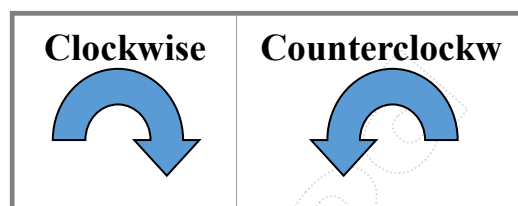
Clockwise CW, Anticlockwise ACW

90° CW, 90°

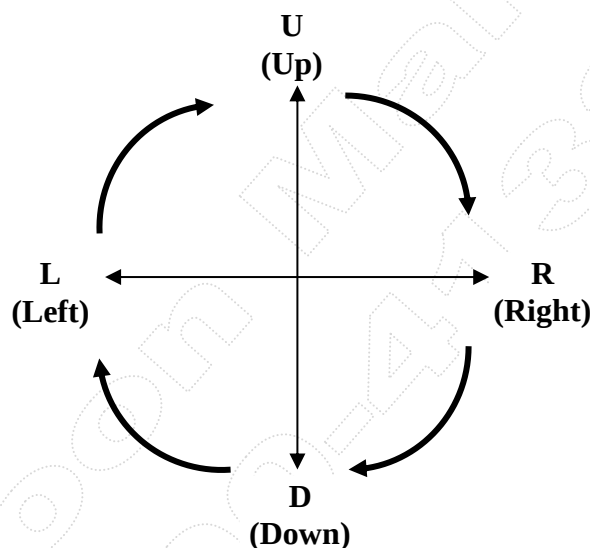
ACW

180° CW, 180° ACW

270° CW, 270° ACW



	Object	Centre	Angle of Rotation	Image
CASE -I	√	√	√	?



Angle of Rotation	90° CW or 270° ACW	270° CW or 90° ACW	180° CW or 180° ACW
	R → D	R → U	R → L
	D → L	U → L	U → D
	L → U	L → D	L → R
	U → R	D → R	D → U

Start always from center and move toward object, Check for each point of the object, measure movement as Right, Left, Up or Down with respect to centre. Then mark every point according to above mention table with respect to centre.

Describe Single Transformation that map A onto B

	Object	Centre	Angle of Rotation	Image
CASE -II	√	?	?	√

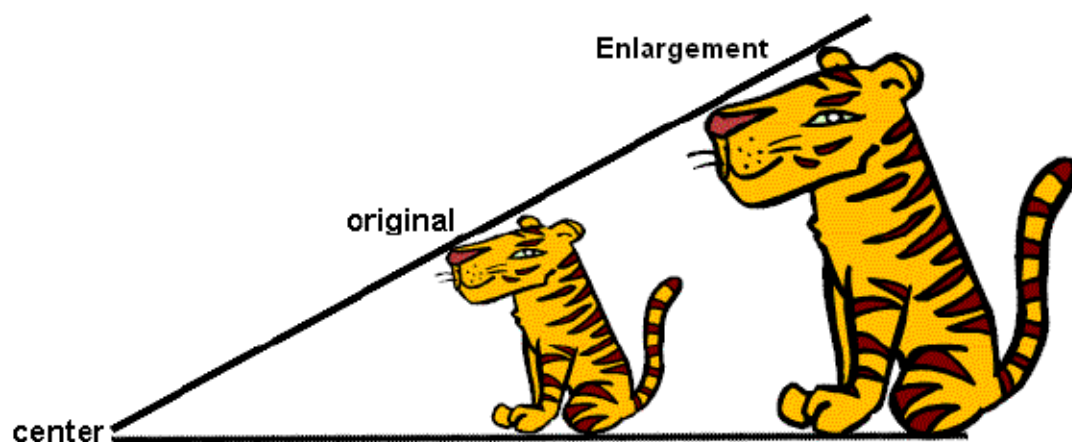
Those transformation in **object** are **Turn or Rotate** to form **Image** are **Rotation**.

To find centre draw **perpendicular bisector** through any two pairs of corresponding points to intersect indicate centre. For Angle, join object and image to centre to measure **angle of Rotation**.

Enlargement

A **Enlargement** (dilation) is a transformation that produces an image that is the **same shape** as the original, but is a **different size**.

A dilation used to create an image **larger** than the original is called an **enlargement**. A dilation used to create an image **smaller** than the original is called a **reduction**.



	Object	Centre	Scale Factor	Image
CASE -I	✓	✓	✓	?

To describe an enlargement state;

- (i) The Scale Factor, K
- (ii) The centre of enlargement

$$\text{Scale Factor} = \frac{\text{Length of the Image}}{\text{Length of the Object}}$$

Enlargement by the scale factors-----centre---

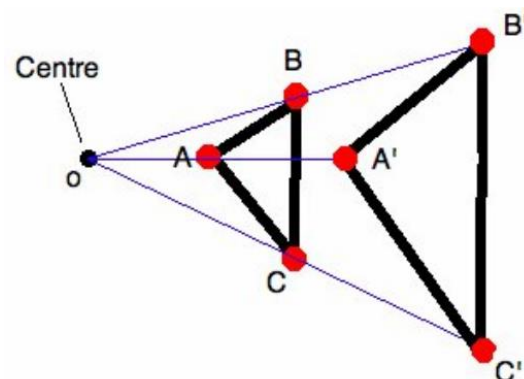
- If $K > 0$, both the object and the image on the same side of the centre of **enlargement**.
- If $K < 0$, the object and the image on the **opposite side of the centre** of enlargement.
- If scale factor lies between 0 and 1, then the resulting **dimension** image is smaller than the object, (although the image is smaller than the object, the transformation it still known as enlargement).

$$\text{Scale Factor} = \frac{\text{Distance between Centre and Image}}{\text{Distance between Centre and Object}}$$

$$\text{Scale Factor} = \frac{\text{Size of Image}}{\text{Size of Object}}$$

$$\text{Area of Image} = (\text{Scale Factor})^2 \times \text{Area of Object}$$

Start always from center and move toward object, Check for each point of object, measure movement as Right, Left, Up or Down with respect to centre. Then multiply each distance with **Scale Factor** and mark every point with respect to centre.



Describe Single Transformation that map A onto B

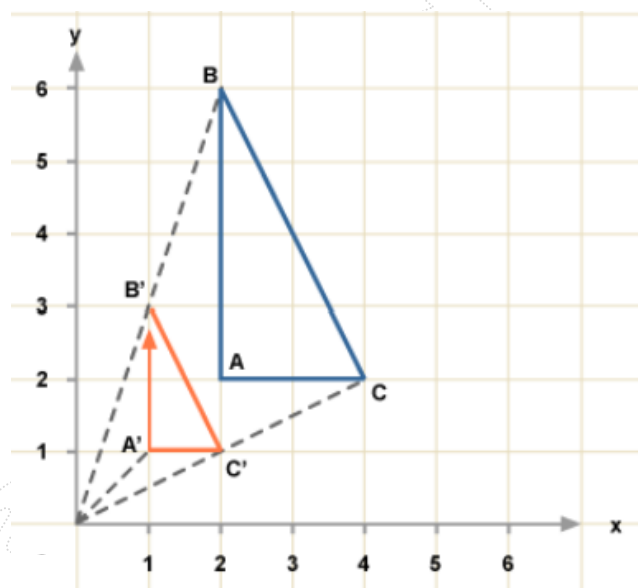
	Object	Centre	Scale Factor	Image
CASE -II	✓	?	?	✓

Those transformation in **object** are **Enlarge or Dimension** to form **Image** are **Enlargement**.

To find centre, join every point object to corresponding point of Image. The point of intersection these lines is centre. For scale Factor use one of following formula

$$\text{Scale Factor} = \frac{\text{Distance between Centre and Image}}{\text{Distance between Centre and Object}}$$

$$\text{Scale Factor} = \frac{\text{Size of Image}}{\text{Size of Object}}$$

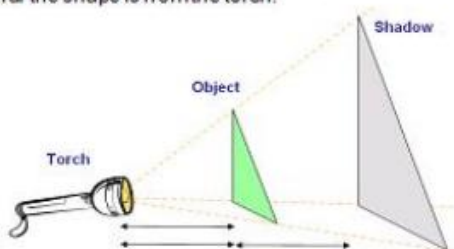


2. Enlargement

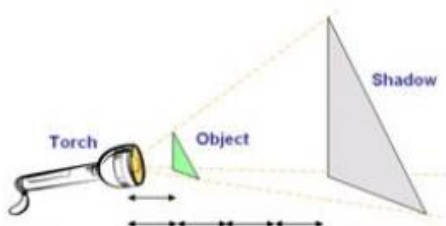
Enlargement is like shining a torch at a shape to make a shadow

The shadow is an enlarged copy of the shape

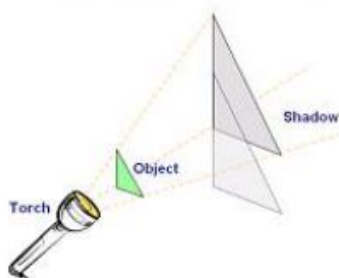
The size of the enlargement depends on how far away the shadow is from the torch compared to how far the shape is from the torch.



Shadow is two times as big as the object since the distance to the shadow is twice as far



Shadow is 4 times as big as the object since the distance to the shadow is 4 times as far



The position of the shadow compared to the object depends on where the torch is placed

Combinations of transformations

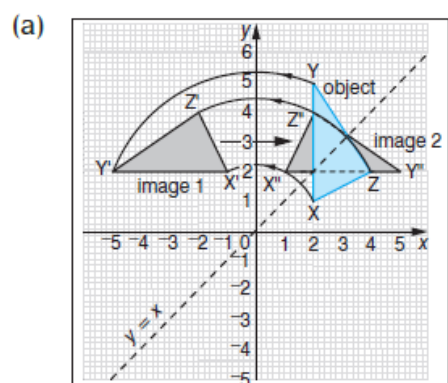
A combination of the same transformation or a combination of different transformations can sometimes be described as a single transformation.

e.g. If R is a reflection in the y -axis and T is a rotation of -90° about the origin, show (on separate diagrams) the image of the triangle XYZ with vertices $X(2, 1)$, $Y(2, 5)$ and $Z(4, 2)$ under these combined transformations:

(a) T followed by R

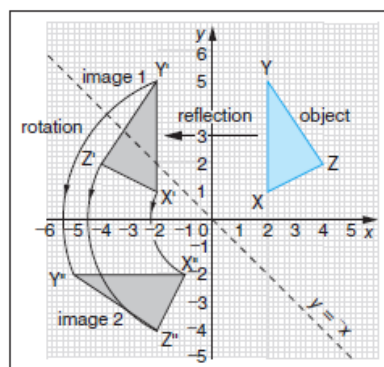
(b) R followed by T.

Which single transformation will return each of these combined transformations back to its original position?



The single transformation that will return $X''Y''Z'$ to XYZ is a reflection in the line $y = x$.

(b)



The single transformation that will return $X''Y''Z''$ to XYZ is a reflection in the line $y = -x$.

NB You can see from the example above that the order of the transformations is important.

Defining transformations

Reflection: define the line of reflection.

Rotation: define the centre of rotation and the angle of rotation (positive or negative).

Enlargement: define the centre of enlargement and the scale factor.

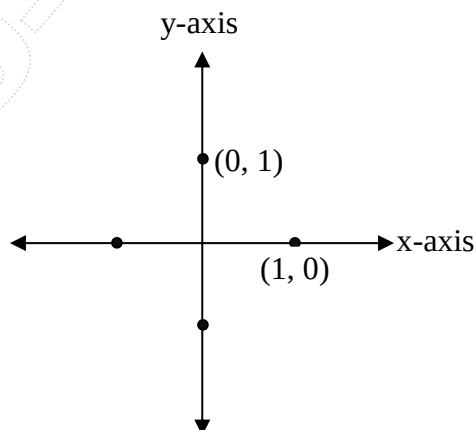
Translation: define the distance and the direction. You can also use vector notation.

Transformation Using Matrices

Transformation	Matrix Equation
Translation $\begin{pmatrix} a \\ b \end{pmatrix}$	$\begin{pmatrix} p' \\ q' \end{pmatrix} = \begin{pmatrix} p \\ q \end{pmatrix} + \begin{pmatrix} a \\ b \end{pmatrix}$
Reflection in y-axis	$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$
Reflection in x-axis	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
Reflection in $y = x$	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
Reflection in $y = -x$	$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$
Clockwise 90° rotation about the origin	$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$
Anticlockwise 90° rotation about the origin	$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$
180° rotation about the origin	$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$
Enlargement with the origin as centre and enlargement factor 'k'	$\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}$

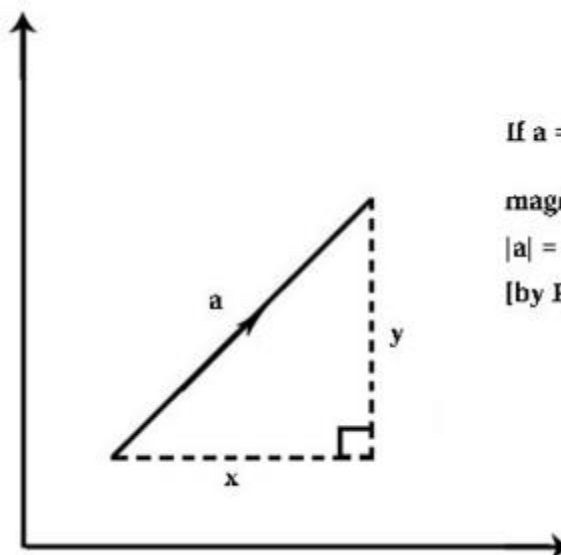
Enlargement

$$\text{Area of Image} = (\text{S.F})^2 \times (\text{Area of object})$$



Topic 30: Vectors in 2D (Two Dimensions)

The magnitude of vector $\overrightarrow{AB}$ is written as $|\overrightarrow{AB}|$.

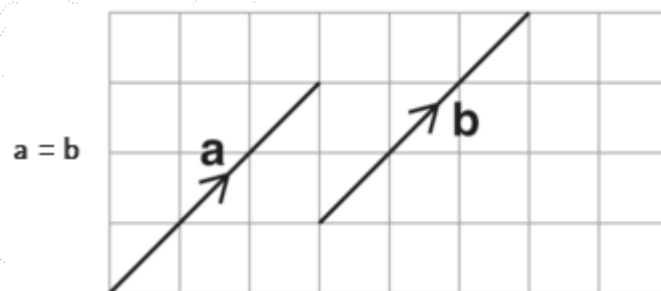


If $\mathbf{a} = \begin{pmatrix} x \\ y \end{pmatrix}$ then the
magnitude
 $|\mathbf{a}| = \sqrt{x^2 + y^2}$
[by Pythagoras Theorem]

Zero vector and unit vectors

A vector with magnitude **0** is called the **zero vector**, written **0**. A vector with magnitude **1** is called a **unit vector**.

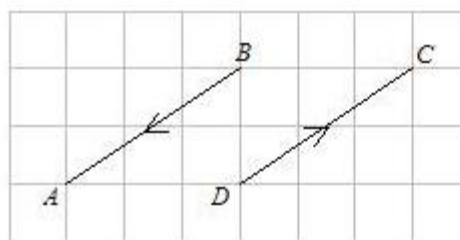
Vectors are equal if they have the same magnitude and the same direction.



Negative Vectors

The negative of a vector is a vector of equal magnitude but in the opposite direction.

A vector $\overrightarrow{AB}$ is a **negative vector** of $\overrightarrow{CD}$, if $\overrightarrow{AB}$ has the same magnitude as $\overrightarrow{CD}$ but are opposite in directions.



$$\overrightarrow{AB} = \begin{pmatrix} 3 \\ 2 \end{pmatrix} \text{ and } \overrightarrow{CD} = \begin{pmatrix} -3 \\ -2 \end{pmatrix} = -\begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

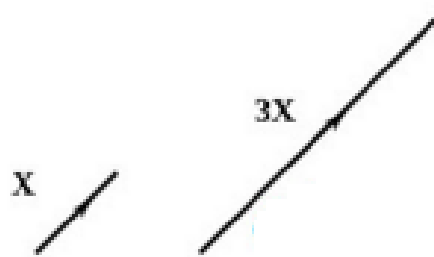
$$\text{So, } \overrightarrow{AB} = -\overrightarrow{CD} \text{ and } -\overrightarrow{AB} = \overrightarrow{CD}$$

$\overrightarrow{AB}$ and $\overrightarrow{CD}$ are negative vectors of each other.

Scalar multiple of vectors

Scalars have magnitude but not direction. Vectors can be multiplied by a scalar to produce another vector.

Multiplying vector x by 3 will give a new vector 3 times the length and parallel to x .



If vector x is $\begin{pmatrix} 3 \\ 3 \end{pmatrix}$ then vector $3x$ is $\begin{pmatrix} 9 \\ 9 \end{pmatrix}$

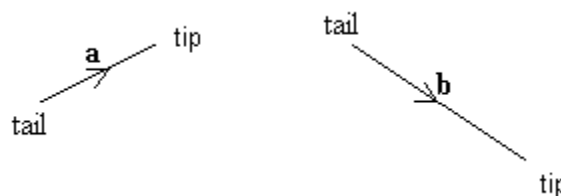
Vector addition and subtraction

When 2 vectors are added or subtracted the vector produced is called the resultant.

The resultant is identified by a double arrowhead.

- add vectors geometrically using the 'Tip-to-tail' method or triangle method
- add vectors using the parallelogram method
- that vector addition is commutative
- that vector addition is associative
- add vectors using components

"Tip-to-Tail" Method



Vectors can be added using the 'Tip-to-tail' method.

Two vectors a and b represented by the line segments can be added by joining the 'tail' of vector b to the 'Tip' of vector a . Alternatively, the 'tail' of vector a can be joined to the 'Tip' of vector b .

Parallel Vectors

Vectors are parallel if they have the same direction. Both components of one vector must be in the same ratio to the corresponding components of the parallel vector.

How to define parallel vectors?

Two vectors are parallel if they are scalar multiples of one another.

If u and v are two non-zero vectors and $u = cv$, then u and v are parallel.

Position Vector

A vector that starts from the origin (O) is called a **position vector**.

In the following diagram, point A has the position vector a and point B has the position vector b .

Find the vector with the help of position vectors of the points

Two points $A(x_1, y_1)$ and $B(x_2, y_2)$

Position vector of A = $OA = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$

Position vector of B = $OB = \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$

Vector $\overrightarrow{AB} = \begin{pmatrix} x_2 - x_1 \\ y_2 - y_1 \end{pmatrix}$

Gradient of Line AB = $m = \frac{y_2 - y_1}{x_2 - x_1}$

If $a = \begin{pmatrix} 3 \\ 7 \end{pmatrix}$, then the coordinates of A will be (3, 7).

Similarly, if $b = \begin{pmatrix} 8 \\ 4 \end{pmatrix}$, then coordinates of B will be (8, 4)

